

Skewness Managed Portfolios

Rui Gong
GMO LLC

John Lynch*
Hofstra University

Richard Ogden†
Rowan University

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Abstract

We show that the returns to many prominent cross-sectional anomalies are driven by a small number of extreme, positively skewed stock returns. Consistent with this, we propose a skewness-managed strategy that predicts ex-ante skewness using firm characteristics and modifies each anomaly portfolio by selecting high-skewness stocks in the long leg and low-skewness stocks in the short leg. Applied to 18 well-known anomalies, the strategy improves returns by an average of 5.45 percentage points, with gains concentrated in periods of economic and financial stress. Sharpe ratios also increase consistently, with an average gain of 0.12. Despite closely tracking the original factors, these skewness-managed portfolios deliver significant alphas even relative to models constructed from the same characteristics, suggesting that modern asset pricing models may systematically overlook higher-order moments.

*Zarb School of Business, Hofstra University, 149 Hofstra University, Hempstead, NY 11549. E-mail: John.Lynch@hofstra.edu.

†Rohrer College of Business, Rowan University, 453 Business Hall, 201 Mullica Hill Road, Glassboro NJ 08028. E-mail: ogdenr@rowan.edu.

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1 Introduction

Aggregate equity returns are primarily driven by a small number of stocks with extreme, positively skewed returns (Bessembinder (2018)). This skewness has important implications for portfolio construction more broadly, since just a few stocks can dominate performance. While this dynamic is well documented at the market level, less is known about its role in long-short anomaly portfolios, which are formed using firm characteristics that do not explicitly account for skewness.¹ In this paper, we explore two related questions: How do positive outliers influence the performance of anomaly portfolios? And can managing exposure to these outliers improve returns?

We begin by showing that a small number of extreme right-tail returns play an important role in driving anomaly performance. We do so by reducing their influence through winsorization, which significantly alters the returns and Sharpe ratios of most anomalies. Motivated by this, we develop a skewness-managed strategy that tilts long-short portfolios toward stocks with high expected skewness in the long leg and away from them in the short leg. Our skewness-managed strategy increases average annual returns by 5.45 percentage points and raises Sharpe ratios by an average of 0.12, with the largest gains observed for Value and Investment.

To assess whether these enhanced returns reflect exposure to new sources of risk, we conduct spanning tests that regress each skewness-managed anomaly on its original counterpart. With the exception of the Beta and Size anomalies, betas are at or below one, while alphas are large and statistically significant, indicating that the performance gains are generally not driven by amplified exposure to the original anomaly. We further test these portfolios against a variety of leading factor models, including models constructed from the same underlying characteristics. In each case, we find consistent pricing failures: the skewness-managed portfolios earn significant excess returns that are unexplained by existing models. These findings imply that skewness represents a dimension of returns that is not captured by standard linear factor models, and that incorporating expected

¹In this paper, we use the term “anomaly” in the empirical asset pricing sense to refer to return patterns that are predictable using firm characteristics, regardless of whether they reflect mispricing or compensation for risk.

skewness into portfolio construction systematically enhances anomaly performance.

Skewness, which captures the asymmetry of a distribution and reflects the likelihood of extreme returns, has long been a topic of interest in finance. Markowitz (1952) famously excluded higher moments like skewness from the mean-variance framework, assuming investors only care about the first two moments. Subsequent work challenged this view by incorporating skewness into asset pricing models (Rubinstein (1973); Kraus and Litzenberger (1976)). Building on this, Harvey and Siddique (2000, 2022) show that stocks with higher coskewness, i.e., greater exposure to aggregate skewness, earn higher expected returns, suggesting that skewness is a priced source of risk. This line of research motivates the treatment of skewness as a risk-based anomaly, akin to other systematic factors.

Complementing the risk-based view is a behavioral perspective in which investors exhibit preferences for positively skewed returns. Models incorporating skewness preferences (e.g., Mitton and Vorkink (2007); Brunnermeier et al. (2007); Barberis and Huang (2008)) predict overpricing of lottery-like stocks, leading to a negative premium for idiosyncratic skewness. This prediction is supported by empirical evidence across a range of methodologies and horizons (e.g., Boyer et al. (2010); Bali et al. (2011); Amaya et al. (2015)). Together, these findings emphasize that skewness, both systematic and idiosyncratic, can be viewed as an anomaly in its own right.

While prior work establishes skewness as a distinct anomaly, relatively little is known about how it interacts with more prominent anomalies. Recent studies suggest that skewness preferences may help explain return patterns in the short leg of some anomalies (Kumar et al. (2019)), proxy for growth options that influence expected returns (Bali et al. (2020)), help account for the alpha in low-risk strategies via coskewness (Schneider et al. (2020)), and document that cross-sectional variation in skewness is persistent and strongly linked to firm characteristics (Albuquerque (2015)). Building on these insights and motivated by Bessembinder (2018), we extend this line of inquiry by exploring whether similarly skewed outcomes drive the performance of well-known anomaly portfolios, and whether managing for skewness can systematically enhance their returns.

We test the role of extreme returns in shaping anomaly performance by comparing annual returns for 18 well-known anomalies to versions constructed using winsorized monthly returns that cap the most extreme right-tail observations.² The 18 anomalies are selected because they are the underlying factors employed in well-established risk- and behavioral-based asset pricing models, including Fama and French (1993) three-factor model [FF3] and Fama and French (2015) five-factor factor model [FF5], Carhart (1997) four-factor model [FFC4], Hou et al. (2015) q-factor model [Q4], the mispricing factors by Stambaugh and Yuan (2017) [SY4], and the behavioral factors in Daniel et al. (2020) [DHS3]. Three caps are applied after portfolio formation, using historical thresholds for the 99th, 95th, and 90th percentiles of the return distribution. The results reveal a clear pattern: the effect of winsorization depends on which leg of the long-short portfolio contains more positively skewed stocks. For example, capping returns tends to reduce performance for strategies like Beta, Size, and Value, where the long leg includes stocks with highly skewed return distributions. In contrast, anomalies such as Momentum, ROE, and ROA see performance gains, consistent with more skewness on the short leg. The skewed stocks driving these effects tend to be those with low profitability, poor past returns, and small market capitalizations — firm characteristics that are associated with high idiosyncratic upside potential. For other strategies, such as Announcement Return, winsorization has little effect, suggesting more symmetric return distributions across legs. Taken together, these findings indicate that skewness plays a central and systematic role in determining anomaly portfolio performance.

Given the impact that skewness has on anomaly portfolios, we propose a skewness-managed strategy that constructs long-short portfolios using information on expected skewness. This approach requires a forward-looking measure of skewness, which we build following Amaya et al. (2015) and Boyer et al. (2010). We first calculate realized skewness from daily returns each month, then estimate monthly cross-sectional regressions of realized skewness on lagged firm characteristics,

²The 18 anomalies we study are: Beta, Size, B/M, Investment, Momentum, Return on Equity, Operating Profitability, Accruals, Announcement Return, Composite Equity Issuance in the past 5 years, Net Stock Issues, Composite Equity Issuance in the past 12 months, Investment to Asset, Failure Probability, OScore, Gross Profitability, Net Operating Assets, and Return on Assets.

such as past returns, momentum, volatility, and industry and exchange indicators. The coefficients from these regressions are then used to construct an expected skewness measure for each stock based on information available at portfolio formation.

Once we have a measure of expected skewness, we implement a sequential sort each month: stocks are first sorted on the anomaly characteristic, and then each leg is divided into terciles based on expected skewness. The skewness-managed strategy takes the long leg from the high expected skewness tercile and the short leg from the low expected skewness tercile. This modification produces substantial gains, with anomaly returns increasing by 5.45 percentage points annually. The largest improvements occur for Value and Investment strategies where returns rise by 9.75 percentage points for Value (B/M) and 9.04 percentage points for Investment (Inv). Profitability-based strategies (Operating Profitability and Gross Profitability) also experience large gains. Sharpe ratios generally rise as well, with an average gain of 0.12. The results remain robust after incorporating transaction costs (Novy-Marx and Velikov (2016)) for a large set of anomalies.

While the performance gains from skewness management are economically large, often doubling the average return of the original strategy, a natural concern is that the gains simply reflect amplified exposure to the original anomaly rather than a genuine improvement. To investigate this, we conduct spanning regressions of the skewness-managed strategies on their corresponding original anomaly portfolios. With the exception of the Beta and Size anomalies, betas are at or below one, yet the managed portfolios still deliver large and statistically significant alphas, typically between five and eleven percentage points annually, indicating that the performance gains are generally not driven by amplified exposure to the original anomaly.

We then examine whether these returns can be explained by modern factor models, including those built from the same underlying firm characteristics. In these regressions, alphas increase across the board relative to the original anomalies, with many previously insignificant alphas becoming large and significant. To understand whether the alpha gains might reflect shifts in factor exposures, we examine changes in betas across both univariate and multivariate settings.

The managed portfolios increase exposure to size-related factors (SMB, ME) and market beta, while decreasing exposure to profitability-related factors (ROE, RMW), consistent with positively skewed returns concentrating in small-cap and low-profitability stocks. Critically, however, these shifts in factor loadings do not account for the large alphas since controlling explicitly for the factors to which loadings change leaves the abnormal performance intact. Taken together, these results suggest that current asset pricing models systematically overlook how return asymmetries shape portfolio performance.

Our main findings are robust to a variety of alternative approaches and specifications. We replicate our strategy using idiosyncratic skewness, the original measure proposed by Boyer et al. (2010), and find similar or even stronger improvements in anomaly performance. We also examine an alternative implementation that scales exposure to each anomaly based on the average expected skewness of portfolio stocks, following a volatility-timing approach similar to Moreira and Muir (2017). While this scaled strategy yields smaller gains in magnitude, the average improvement in returns remains statistically significant. In additional tests, we verify that our results are not driven by known return predictors used in constructing the expected skewness measure. Specifically, we find that the skewness-managed strategies remain effective even when these predictors are excluded from the estimation process. We also demonstrate that a long-short portfolio formed directly on expected skewness does not fully explain the returns of the skewness-managed strategies, indicating that our approach captures more than just the predictive content of skewness itself.

Beyond the cross-sectional evidence, we examine whether the return premium generated by skewness management varies over time and whether it is related to macroeconomic conditions. To do so, we construct the equal-weighted average increment, defined as the average monthly return difference between the skewness-managed and standard versions of each anomaly across our 18 strategies. While this increment is positive in only 54% of months, the strategy nonetheless generates large cumulative outperformance, with a \$1 investment in the equal-weighted skewness-managed portfolio growing to approximately \$530 by December 2024, compared to \$22 for the

standard portfolio. This implies that the gains in positive months substantially exceed the losses in negative months rather than reflecting consistent monthly improvement. This episodic pattern motivates a closer examination of macroeconomic conditions. We find that the increment is highly cyclical, averaging 20.4% per year during NBER recessions compared to 3.7% during expansions. Pooled time-series regressions confirm this pattern, with the nonlinear real activity factor $F1^3$ of Ludvigson and Ng (2007) and the VIX emerging as the strongest predictors. Together with excess market return, these variables explain up to 9.3% of monthly variation. At the individual anomaly level, the conditional alphas are higher in stressed states for virtually all 18 anomalies, while their conditional loadings on the original factors vary in both sign and magnitude. Importantly, the premium is not exclusively a bad-times phenomenon, as the strategy generally generates significant alphas across all 18 anomalies even when macro conditions are at their average levels.

This paper contributes to several strands of literature. As discussed earlier, skewness has long played a role in asset pricing. Rubinstein (1973) and Kraus and Litzenberger (1976) extended the CAPM to incorporate coskewness, showing that stocks with greater exposure to systematic skewness earn higher expected returns, a result confirmed in both early and modern data (Harvey and Siddique (2000, 2022)). Subsequent work has refined our understanding of systematic skewness. Adrian and Rosenberg (2008) document a positive price of market skewness risk, while Chang et al. (2013) use daily S&P 500 index option data to show that stocks with high sensitivities to innovations in implied market volatility and skewness earn lower returns. Langlois (2020) decomposes individual stock skewness into systematic and idiosyncratic components, finding that the expected systematic skewness factor commands a significant premium. Neuberger (2012) and Neuberger (2022) provide complementary evidence that skewness is pervasive, horizon-dependent, and can be robustly measured using realized high-frequency data. Karehnke (2024) further shows that systematic skewness carries time-varying pricing implications. Together, this literature highlights systematic skewness as an important state variable in asset pricing, with evidence of economically meaningful risk-based pricing across different settings.

A parallel literature investigates the role of investor skewness preferences in driving cross-sectional returns. Models in this tradition argue that investors have a preference for lottery-like, positively skewed payoffs, leading to systematic overpricing of such stocks and a negative idiosyncratic skewness premium (Mitton and Vorkink (2007); Brunnermeier et al. (2007); Barberis and Huang (2008)). This prediction is supported by empirical evidence across multiple settings (Boyer et al. (2010); Bali et al. (2011); Conrad et al. (2014); Amaya et al. (2015)). More recent work links skewness preferences to anomaly returns, suggesting that they may drive overpricing in the short leg (Kumar et al. (2019)) or proxy for embedded growth options (Bali et al. (2020)). Our paper takes a broader perspective by examining the general interaction between skewness and well-known anomalies, showing that a relatively small number of high-skewness stocks can disproportionately drive anomaly returns.

Our paper also contributes to the growing body of work on the limitations of modern factor models. Recent studies highlight that even the most prominent models struggle to price test assets derived from their own underlying characteristics. Baba Yara et al. (2022) highlight how no single model is able to fully price the cross-section of stock returns despite the weak theoretical assumptions needed for one to exist. This limitation extends to portfolios constructed from the models' own factors. Baba Yara et al. (2021) find that when portfolios are formed using older sorts of the same characteristic, especially with longer holding periods, factor models often fail to explain returns, particularly for models with a greater number of factors. Similarly, Chernov et al. (2022) demonstrate that factor models fail to price their own factors when evaluated using multihorizon returns as test assets. Relatedly, Bianchi et al. (2021) show that time-varying skewness plays an important role in explaining momentum crashes, highlighting the importance of higher-moment dynamics in factor returns. Our results complement this line of research by showing that skewness-managed portfolios, created from the same underlying characteristics used to construct the factors, consistently earn large and significant alphas relative to the models they are meant to validate. In this way, we provide new evidence that these models are not only incomplete but are systematically

unable to capture dimensions of return behavior related to skewness.

We also build on a parallel literature that examines how return distribution features can be actively managed. A close analog is the literature on volatility-managed portfolios. For instance, Barroso and Santa-Clara (2015) and Daniel and Moskowitz (2016) show that scaling momentum strategies by inverse volatility reduces crash risk and improves performance. Moreira and Muir (2017) generalize this approach by demonstrating that volatility management improves Sharpe ratios across a range of equity factors, often generating alpha relative to the original strategies. However, follow-up studies such as Barroso and Detzel (2021) raise concerns that turnover and trading costs may offset these gains, while Cederburg et al. (2020) argue that improvements in alpha are not always accompanied by improvements in risk-adjusted returns. Our paper contributes to this literature by proposing a novel skewness-based management approach that, like volatility scaling, is simple to implement but offers significant improvements in both returns and Sharpe ratios, with limited added exposure to traditional risk factors. Importantly, our method operates on the cross-section of individual stocks, capturing return asymmetries that are missed by time-series volatility alone.

Finally, we are inspired by and build upon the insights of Bessembinder (2018), who demonstrates that a small number of stocks account for the entire net gain in the U.S. equity market over long horizons. His findings highlight the importance of positive outliers in equity performance, especially when analyzing buy-and-hold portfolios. Our work extends this insight to long-short strategies based on firm characteristics, showing that positive skewness is not just a market-level phenomenon but a critical feature of anomalies themselves. We document that many anomalies rely disproportionately on a small subset of stocks with extreme right-tail returns. While these outliers tend to be concentrated in one leg of the portfolio, typically the long leg for Value and Size strategies, or the short leg for profitability anomalies, our skewness management strategy does not rely on identifying this concentration in advance. Instead, it systematically tilts each leg of a given anomaly portfolio based on the cross-sectional distribution of expected skewness,

enhancing performance regardless of where the skewness resides. This contributes to the broader understanding of how skewness influences equity returns, not only in aggregate but also within the structure of characteristic-based investment strategies.

2 Motivation

In this section, we examine the role of extreme returns in driving anomaly premiums by evaluating the performance of common anomalies in a winsorized sample of stock returns.

2.1 Data

We draw our sample from the Center for Research in Security Prices (CRSP) universe of common stocks, focusing on the period from July 1963 to December 2024. The 18 anomaly strategies are constructed using data from CRSP, Compustat, and the Open Source Asset Pricing database of Chen and Zimmerman (2022).³ The 18 anomalies are Beta, Size, Book-to-Market, Investment, Momentum, Return on Equity, Operating Profitability, Expected Growth, Accruals, Announcement Return, Composite Equity Issuance in the past 5 years, Net Stock Issues, Composite Equity Issuance in the past 12 months, Investment to Asset, Failure Probability, OScore, Gross Profitability, Net Operating Assets, and Return on Assets. Further details are provided in the Appendix.

To identify extreme-return observations, we calculate monthly percentiles from the historical distribution of returns for each month t , using all observations from January 1926 up to month t . These historical thresholds allow us to define the “tail” of the distribution for each month in a way that reflects the long-run properties of stock returns. Our results are robust to alternative definitions of the percentiles based on the full-sample distribution.

2.2 The Role of Skewness in Equity Returns

How skewed is the return distribution? Panel A of Figure 2 shows a histogram of monthly returns from June 1963 to December 2024. The distribution exhibits a long right tail — barely visible at

³<https://www.openassetpricing.com/>

this scale — that extends to an extraordinary 2,658% monthly return. This right tail contributes to substantial asymmetry, with a sample skewness of 9.06. These extreme observations are not evenly distributed over time but instead cluster in specific periods. Panel B of Figure 2 plots the percentage of stocks exceeding a return threshold of 26.8% (approximately the top 5% of monthly observations). Many of these large positive returns occur during sharp market rebounds following downturns, such as the periods following the tech bust, the global financial crisis, and the COVID-19 crash.

The existing literature documents that positive skewness in stock returns is an important driver of long-run equity performance, with a small number of stocks generating a disproportionate share of total market wealth creation (Bessembinder (2018)). In the context of long-short anomaly portfolios, this dynamic implies that a relatively small set of high-performing stocks can have an outsized effect on portfolio returns. Whether these extreme observations help or hurt investors depends on how they are distributed across the long and short legs of the portfolio.

Strategies whose long legs hold stocks with highly skewed return distributions tend to benefit from the presence of outliers. In contrast, strategies whose short legs contain more positively skewed stocks tend to suffer, as these short positions are occasionally exposed to large upward price moves. Some characteristics, however, are not systematically related to skewness in individual stock returns, implying that extreme right-tail outcomes have little systematic effect on the strategy’s performance.

To quantify the influence of these outliers, we conduct a simple experiment: we reconstruct each anomaly portfolio from a winsorized sample in which extreme positive returns are capped ex post. This capped sample approximates an investment universe in which the most extreme right-tail outcomes are not attainable, allowing us to observe how anomaly returns behave in their absence. The return caps are chosen from the monthly cross-sectional return distribution using three thresholds: the 90th, 95th, and 99th percentiles. The winsorization is applied after stocks are sorted into portfolios based on the anomaly characteristic, but before computing the long-short

portfolio return.

Table 1 reports the results for a representative set of anomalies. The effects of winsorization vary systematically depending on whether skewness is concentrated in the long or short leg. For example, anomalies based on size and book-to-market ratios tend to be negatively affected, as capping returns removes large positive outliers primarily from the long leg, consistent with the tendency of small and growth stocks to exhibit high positive skewness. By contrast, anomalies based on profitability and momentum tend to benefit from capping extreme right-tail returns, consistent with the short leg containing more highly skewed stocks (such as low-profitability firms and past losers). As a neutral example, investment-based anomalies show little change, suggesting that skewness is relatively symmetric across both legs.

From this analysis, we draw three conclusions that motivate our skewness-management strategy. First, skewness can have asymmetric effects on anomaly performance, with some strategies benefiting from extreme returns and others being harmed. Second, although skewness effects tend to be concentrated in either the long or short leg, an investor will generally prefer to take long positions in stocks drawn from more positively skewed distributions and short positions in stocks from less skewed distributions. Third, the stocks most likely to generate large positive outliers are associated with identifiable firm characteristics available at portfolio formation. These insights form the basis for our skewness-management approach developed in the following sections.

3 Predicting Skewness

We construct a measure of *expected skewness* to identify stocks that are more likely to exhibit extreme returns in the following month. This measure is based on a model of *realized skewness*, which we estimate each month from daily returns, and then project onto firm characteristics to obtain an ex-ante forecast.

For each firm i in month t , realized skewness is computed as:

$$rs_{i,t} = \frac{1}{N(t)} \frac{\sum_{d=1}^{N(t)} r_{i,d}^3}{rv_{i,t}^3},$$

where realized volatility is:

$$rv_{i,t} = \left(\frac{1}{N(t)} \sum_{d=1}^{N(t)} r_{i,d}^2 \right)^{1/2}.$$

Here, $r_{i,d}$ is the daily return of stock i on day d , and $N(t)$ is the total number of trading days in month t .⁴

Table 2 shows the summary statistics of Skewness, Volatility, Momentum, prior return, and turnover. Unsurprisingly, this contemporaneous measure of skewness captures the majority of extreme-return events. Approximately 97% of monthly return observations in the positive tail of the distribution lie above the median realized skewness,⁵ while about 97% of left-tail observations fall below it.

The second step in constructing expected skewness is to model realized skewness using lagged firm characteristics. Unlike volatility, which exhibits persistence and can often be proxied with its own lag, skewness is only weakly persistent. Consequently, an ex-ante forecast of skewness requires modeling its cross-sectional relation to firm characteristics. We follow the general approach of Boyer et al. (2010) and estimate monthly cross-sectional regressions using firm characteristics as predictors. However, our measure of skewness differs from theirs. Specifically, we use realized raw return skewness as the dependent variable, whereas Boyer et al. (2010) focus on predicting idiosyncratic skewness obtained from residual returns. While the set of explanatory variables is similar, the difference in the skewness measure implies that our forecasts capture variation in total return skewness rather than idiosyncratic skewness. Consistent with this distinction, our estimated expected skewness measure is associated with a return premium rather than the skewness discount

⁴ $N(t)$ is adjusted for degrees of freedom: one for volatility and two for skewness.

⁵The positive tail is defined as those monthly returns above the historical 90th percentile.

documented in Boyer et al. (2010). We estimate monthly cross-sectional regressions of the form:

$$rs_{i,t} = \beta_t + \lambda_t' X_{i,t-1} + \epsilon_{i,t},$$

where $X_{i,t-1}$ is a vector of firm characteristics measured at the end of month $t-1$ and $rs_{i,t}$ is realized skewness in month t . Running the regression separately each month allows the relationship between characteristics and skewness to vary over time.

We then use the estimated coefficients to generate an ex-ante forecast for month $t+1$:

$$E[rs_{i,t+1}] = \beta_t + \lambda_t' X_{i,t}.$$

The set of firm characteristics X is chosen based on prior literature linking them to skewness. These include realized skewness (RS) and realized volatility (RV) from equations (1) and (2), momentum (measured as the cumulative return over the past year excluding the most recent month), prior-month return, turnover, and indicator variables for size, Fama–French 48 industry classification⁶, and Nasdaq listing. Table 2 reports summary statistics for these variables; Panel A provides descriptive statistics and Panel B reports correlations among the continuous variables.

In line with prior literature, we use a parsimonious specification to maximize the available sample. The final model excludes turnover, which is only available for Nasdaq stocks after 1988, and omits book-to-market to avoid unnecessary data loss.⁷

Table 3 reports results from the monthly cross-sectional regressions. Each row corresponds to a different specification, and coefficients are averaged across all monthly regressions. We also report the percentage of months in which each coefficient is significant at the 10% level and has the same sign as the overall average ($\%sig$). Two predictors emerge as particularly consistent: realized volatility and prior-month return. Consistent with earlier findings, a one-standard-deviation increase in volatility has roughly twice the impact on future skewness as a similar increase in

⁶See Fama and French (1997) and Kenneth French’s data library.

⁷Book-to-market has been found to be a predictor of skewness in prior work (e.g., Chen et al. (2001)), but we prioritize maximizing the overlap with our anomaly samples.

realized skewness. Similarly, higher prior-month returns predict higher expected skewness, with an effect size nearly twice that of volatility. We use model 7 throughout the remainder of the paper. The model’s average R^2 of 3.0% is comparable to other short-horizon skewness prediction models in the literature. Panel A of Table A1 reports the cross-sectional distribution of the stock-level skewness forecasts.

This expected skewness measure forms the basis of our skewness-management strategy described in the following section.

4 Skewness-Managed Portfolios

We use the expected skewness measure outlined in Section 3 to enhance the performance of a broad set of anomaly portfolios. We selected 18 anomalies that are widely studied in the literature and are drawn from the set of firm characteristics used in the construction of several prominent factor models. The summary statistics of monthly returns of the anomalies are reported in Panel B of Table A1. Our skewness-management strategy applies a second sequential sort to each anomaly portfolio, splitting stocks into high- and low-expected skewness groups. The strategy takes the long leg of the anomaly from the high-expected skewness group and the short leg from the low-expected skewness group. This design reflects the intuition that an investor would want to seek stocks with greater potential for extreme positive returns in the long leg, while avoiding them in the short leg.

4.1 Sequential Sorts on Expected Skewness

For each month, we first sort all common stocks in CRSP into deciles based on the relevant anomaly characteristic, following the portfolio definitions described in Appendix A. Within each of the long and short legs, we then sort stocks into terciles based on expected skewness. This produces 30 double-sorted portfolios (10 characteristic deciles $\times$ 3 expected skewness terciles). The returns of these portfolios are computed as value-weighted averages using lagged market capitalization.

Table 4 reports the average returns for these 30 portfolios for six representative anomalies.

The $H-L$ columns show the return spread between the high- and low-expected skewness portfolios within each characteristic decile, while the $L-S$ rows report the difference between the long and short sides of the strategy.

A consistent pattern emerges across all six anomalies: the $H-L$ spread is positive in every characteristic decile. This means that, regardless of the characteristic level, portfolios composed of high-expected-skewness stocks outperform those with low expected skewness. This finding contrasts with the negative expected skewness premium documented in Boyer et al. (2010). One important distinction is that our expected skewness measure is constructed from forecasts of realized raw return skewness, whereas Boyer et al. (2010) focus on predicting idiosyncratic skewness based on residual returns. As a result, our measure captures variation in total return asymmetry rather than idiosyncratic lottery-like payoffs, which may help explain why we observe a positive rather than negative skewness premium. The $H-L$ spread is positive in every decile across anomalies. As we discuss next, however, the magnitude of this spread depends on whether skewness is concentrated in the long or short leg of the anomaly.

For anomalies in Panel A, such as Book-to-Market (B/M), Size, and Investment, the $H-L$ spread is largest in the long leg, reflecting the greater prevalence of skewed stocks among long positions. For example, in the top B/M decile, the spread between high- and low-skewness portfolios is 12.14% annually, compared to just 2.89% in the short leg. This pattern is consistent with high B/M (value) stocks in the long leg often being small-cap firms with substantial upside potential, whereas low B/M (growth) stocks in the short leg tend to have more symmetric return distributions.

In contrast, anomalies in Panel B, such as Momentum, Return on Equity (ROE), and Operating Profitability (OP), exhibit their largest $H-L$ spreads in the short leg, indicating that skewness is concentrated among the stocks being shorted. For instance, in the short leg of ROE, the spread between high- and low-skewness portfolios is 8.99% annually, versus 2.05% in the long leg. This is consistent with short positions often containing distressed or low-quality firms that can experience large, sudden rebounds, generating positively skewed returns.

Importantly, the high-expected-skewness portfolios in each tercile sort also contain stocks with materially higher *realized* skewness on average, confirming that the expected skewness measure successfully identifies the firms most likely to experience extreme positive returns. As reported in Table A2 in the Appendix, the difference in realized skewness between the high- and low-expected-skewness portfolios ($H-L$) is positive for every characteristic decile across all six representative anomalies. Moreover, the largest $H-L$ differences occur in the portfolio leg where skewness is most concentrated — for example, the long leg for B/M, Size, and Investment, and the short leg for Momentum, ROE, and OP — mirroring the return spread patterns observed in Table 4. This alignment between expected and realized skewness underscores that the measure is capturing genuine cross-sectional variation in return asymmetry, rather than noise.

These results reinforce two key points from our earlier winsorization analysis. First, skewness is a systematic and predictable feature of anomaly portfolios, linked to observable firm characteristics at portfolio formation. Second, the economic benefit from sorting on expected skewness is greatest in the portfolio leg where high-skewness stocks are concentrated — typically the long leg for value and size strategies, and the short leg for Momentum and profitability strategies.

4.2 Performance of the Skewness-Managed Strategy

Table 5 reports the performance of the skewness-managed strategies for all 18 anomalies in our sample. Panel A presents the mean returns, Sharpe ratios, and return skewness for the standard anomaly portfolios, calculated in the subsample with non-missing expected skewness. Panel B reports the same statistics for the skewness-managed versions, where the long leg is drawn from the high expected skewness tercile and the short leg from the low expected skewness tercile. Panel C shows the difference between the two portfolios. Newey-West t -statistics for the mean return differences, t -statistics for the skewness differences, and Jobson and Korkie (1981) z -scores for the Sharpe ratio differences are reported in parentheses.

The results show that skewness management improves the performance of every anomaly

strategy considered. Mean annual returns increase substantially, with gains ranging from 0.65 percentage points for Net Operating Assets (NOA) to 9.04 percentage points for Value (B/M). The improvements are most striking for Value and Investment: for the Value (B/M) anomaly, the average annual return rises from roughly 3.4% in the standard portfolio to over 13% in the skewness-managed version — nearly a four times increase. For Investment, the mean return more than doubles, increasing by about 9 percentage points. The average improvement across anomalies is approximately 5.45 percentage points per year. Sharpe ratios also rise for most anomalies, with notable gains for B/M (0.32), Operating Profitability (0.32), and Composite Equity Issuance of the past 12 months (0.44). The improvements in Sharpe ratios are statistically significant for several anomalies, including Beta, Size, B/M, Operating Profitability, Composite Equity Issuance of the past 12 months, and Gross Profitability. Even for anomalies where Sharpe ratio changes are modest or negative, such as Announcement Return (AnnRet), average returns still increase meaningfully. In addition, the realized return skewness of the managed portfolios is meaningfully higher than that of the standard anomalies for many factors.

These results indicate that the performance gains from skewness management are broad-based, occurring regardless of whether skewness is concentrated in the long leg, the short leg, or more evenly distributed across both (as with Investment). The strategy’s design of tilting toward high expected skewness in the long leg and away from it in the short leg systematically captures the upside potential of positively skewed stocks while mitigating adverse effects from holding them on the short side. In many cases, such as Size, Value, and Gross Profitability, skewness management more than triples the mean return relative to the standard anomaly. Taken together, the evidence in Table 5 demonstrates that a simple, transparent adjustment to portfolio construction informed by expected skewness consistently enhances the performance of a wide range of well-known anomalies.

We also explore a dynamic scaling implementation of skewness management, in which portfolio weights are adjusted proportionally to the average expected skewness in each leg. More details are discussed in 4.4. While the qualitative improvements in performance remain, the magnitude of

gains is smaller. These results are reported in Table A3 in the Appendix.

4.3 Factor Model Tests and Implications

To evaluate whether the enhanced returns of the skewness-managed strategies can be explained by existing risk models, we estimate spanning regressions and factor model alphas. The factor models we test include: Fama and French (1993) [FF3]; Carhart (1997) [FFC4]; Fama and French (2015) [FF5]; Hou et al. (2015) [Q4]; Stambaugh and Yuan (2017) [SY4]; and Daniel et al. (2020) [DHS3].

Table 6 reports the results from spanning regressions in which each skewness-managed anomaly is regressed on its corresponding standard anomaly portfolio. The intercepts are economically large: the annualized alpha averages roughly 6.7% across the 18 anomalies, ranges from about 2.6% to 10.8%, and is statistically significant at the 5% level for 14 of the 18 cases. The loadings on the standard anomaly are, with two exceptions, at or below one — averaging about 0.8 across the 18 anomalies — implying that for most factors the managed portfolio attains higher returns while taking on the same or less than full exposure to the original anomaly. The two exceptions are the Size and Beta anomalies, for which the managed portfolio has a loading on the standard anomaly that exceeds one. This pattern is intuitive: stocks with high expected skewness are disproportionately concentrated among small and high-beta stocks (Bali et al. (2011)), so tilting the long leg of each anomaly toward high-expected-skewness stocks mechanically increases its exposure to small-cap and high-beta segments of the cross-section. Taken together, the positive alphas alongside loadings that are typically equal or below one indicate that the standard anomaly does not span the skewness-managed portfolio: its return cannot be replicated by simply scaling exposure to the original strategy. The performance gains documented in Section 4.2 therefore represent an incremental source of return that is not captured by the corresponding standard anomaly.

Table 7 compares the alphas from the skewness-managed strategies to those from the original anomalies across the seven factor models. In nearly every case, alpha increases substantially, with some anomalies shifting from statistically insignificant or marginally significant alphas to

large, highly significant ones. Figure 1 visualizes the comparisons. Consider B/M anomaly, it is completely subsumed by the Fama-French five-factor model, as expected. In contrast, the skewness-managed B/M strategy delivers an alpha of 6.93% that remains unexplained by the same model. These results underscore a central point: even after controlling for a wide range of systematic risk factors, including those constructed from the same characteristics as the anomalies themselves, the skewness-managed strategies continue to deliver sizable abnormal returns. This indicates that current factor models fail to capture the return dimension associated with skewness management.

Tables 8 and 9 examine how skewness management alters factor exposures in both univariate and multivariate settings. Consistent patterns emerge: the managed portfolios tend to have higher loadings on size-related factors (SMB, ME) and higher market betas, while loadings on profitability-related factors (ROE, RMW) generally decline. Some anomalies also show reduced exposure to investment-related factors (IA, CMA). These shifts indicate modest but systematic changes in portfolio composition and risk profile following skewness management. However, the key takeaway is that these changes in factor loadings do not account for the large and significant alphas observed in Tables 6 and 7. In other words, even when we explicitly control for the factors to which loadings shift, substantial abnormal performance remains.

Overall, these results reinforce that skewness-managed strategies are not simply repackaged versions of the original anomalies with altered exposures to known risk factors. Rather, they capture a return component tied to the interaction between skewness and the underlying anomaly characteristic — a dimension of the cross-section of returns that current linear factor models leave unexplained.

4.4 A Variant of Skewness-Managed Strategy - Dynamic Scaling

We also examine a dynamic scaling strategy that uses our expected-skewness signal not to choose which stocks to hold, but to scale the size of the standard long-short anomaly portfolio over time. Concretely, for each anomaly we keep the standard long-short decile portfolio (decile 10 minus

decile 1) intact, and we modulate its month-by-month size using a single time-series signal s_t defined as the average of (a) the mean expected skewness in the long-leg decile in month t and (b) the mean expected skewness in the short-leg decile in month t . The signal therefore measures how high expected skewness is at the cross-sectional extremes in a given month. We then rank s_t against its own expanding-window history to obtain a real-time percentile $p_t \in [0, 1]$, using only information available up to time t . We consider two scaling rules: a “Long” rule that multiplies the standard long–short return by $2p_t$ — so the bet is larger when expected skewness at the extremes is unusually *high* relative to its own history — and a “Short” rule that multiplies it by $2(1 - p_t)$ — so the bet is larger when expected skewness at the extremes is unusually *low*. The “Long”/“Short” labels thus describe the direction of the timing rule, not differential treatment of the long and short legs: in both rules the long and short legs are scaled by the same factor, so only the gross size of the long–short bet varies. As a numerical example, when the signal is at the 75th percentile of its historical distribution, the “Long” rule takes $2 \times 0.75 = 1.5$ times the standard long–short exposure that month, while the “Short” rule takes $2 \times (1 - 0.75) = 0.5$ times. Because the scaling factor lies in $[0, 2]$ with mean one, this strategy preserves the gross long–short exposure of the standard anomaly and only reallocates it over time. For each anomaly we report whichever of the two rules produces the higher Sharpe ratio, together with the standard unscaled long–short portfolio for comparison.

The results, reported in Table A3 in the Appendix, show that the qualitative improvement in performance documented in Table 5 largely survives this purely time-series form of skewness management. For roughly half of the anomalies, the scaled strategy delivers significantly higher annualized mean returns. The magnitude of these gains, however, is smaller than that produced by the cross-sectional strategy in Table 5. On average across the 18 anomalies, dynamic timing skewness management improves the mean return of a factor by approximately 2.46 percentage points per year.

5 Robustness

We conduct several robustness checks to ensure that our main results are not driven by the construction of our expected skewness measure or by the particular choice of realized skewness as the underlying target variable. More specifically, we address the following concerns: (1) that the outperformance of skewness-managed portfolios disappear after transaction costs; (2) that performance gains simply reflect the standalone effect to expected skewness; (3) that our findings hinge on the particular definition of skewness we use; and (4) that our results depend on specific predictors included in the expected skewness model.

5.1 Performance Net of Transaction Costs

To evaluate the implementability of our strategy, we compute portfolio returns after accounting for trading costs. We follow the portfolio-level transaction costs methodology of Novy-Marx and Velikov (2016) and use stock-level estimates of effective spreads from Hasbrouck (2009), which are based on a Gibbs-sampling implementation of the generalized Roll (1984) model. However, several recent contributions argue that this measure of costs substantially overstates the true execution costs faced by sophisticated institutional traders. Frazzini et al. (2018) find actual trading costs to be an order of magnitude smaller than previous studies suggest. We therefore view our transaction-cost-adjusted results as a conservative lower bound on the economic significance of the skewness-managed strategies; the true after-cost performance for an institutional investor is likely to be meaningfully better than what we report.

Following Hasbrouck (2009), the resulting measure, denoted $c_{i,t}$, represents the estimated one-way trading cost of stock i in month t as a fraction of price. Table A4 shows a summary statistics of transaction costs estimates. On average, 2% of the market capitalization has missing transaction costs estimates. The number has declined since the decimalization era. In addition, to model trading behavior of institutional investors, We exclude month-stock observations if the transaction costs estimate is higher than 0.01. This is a realistic assumption given that asset managers and

institutional investors typically exclude stocks with extremely high estimated transaction costs. These observations make up for less than 2% of the total market capitalization.⁸

Transaction costs arise from rebalancing trades. Let $w_{i,t}^{\text{target}}$ denote the target portfolio weight of stock i at the beginning of month t , constructed using information available at the end of month $t - 1$. Let $w_{i,t}^{\text{after}}$ denote the weight implied by the previous month's holdings after accounting for realized returns. The required trade is therefore

$$\Delta w_{i,t} = w_{i,t}^{\text{target}} - w_{i,t}^{\text{after}}.$$

Following Novy-Marx and Velikov (2016), the transaction costs incurred in month t is computed as

$$TC_t = \sum_i c_{i,t} |\Delta w_{i,t}|.$$

For long-short portfolios, transaction costs are computed separately for the long and short legs and aggregated to obtain the total cost of rebalancing. Net portfolio returns are then given by

$$R_t^{\text{net}} = R_t^{\text{gross}} - TC_t,$$

where R_t^{gross} denotes the gross portfolio return prior to trading costs.

The results of portfolio performance comparisons are shown in Table A5 in Appendix. The performance gains of Size, B/M, and Investment remain significant after incorporating transaction costs. And the performance gains of Beta and 12-month Composite Equity Issuance remain marginally significant. On average, the strategy improves annual returns by 1.58 percentage points across factors. Nevertheless, the magnitude of the return improvement is attenuated once the transaction costs are taken into account. This is expected because the skewness-managed portfolios exhibit higher turnover than the original factor portfolios.

⁸The effective spread estimates are obtained from the transaction costs data library maintained by Joel Hasbrouck, available on his website. The effective spread estimates post 2009 are calculated using the code provided by Hasbrouck on his website. The estimates are based on the Gibbs-sampling implementation of the generalized Roll model described in Hasbrouck (2009).

5.2 Spanning Tests with Expected Skewness Factor

Another concern is that the skewness-managed anomalies may simply be capturing the well-known anomaly for holding high-expected-skewness stocks. To address this, we construct a long-short factor that is long stocks in the highest expected-skewness tercile and short stocks in the lowest tercile, and include it as an additional explanatory variable in spanning tests alongside the original anomaly and the Fama-French five-factor model.

Table A6 reports the spanning-test results. The skewness-managed strategies are not absorbed by the standalone $E[Skew]$ anomaly: in Panel A, all but two of the 18 skewness-managed long-short returns (Beta and Size are the exceptions) generate statistically significant alphas with respect to $E[Skew]$ alone, with annualized point estimates ranging from 1.9% (Size) to 17.4% (momentum). This pattern is not surprising in light of the asset-pricing literature on skewness preference, which generally documents a *discount* rather than a *premium* associated with high expected skewness (e.g., Boyer et al. (2010); Bali et al. (2011); Conrad et al. (2013)). In our specification, the unconditional $E[Skew]$ long-short does earn a positive average return, but the magnitude is economically modest (2.64% per year, $t = 1.66$) and statistically insignificant at the 5% level. We attribute this positive sign to the composition of our prediction model, which loads positively on the small-cap dummy (a characteristic that itself commands a positive premium in our sample) and negatively on the prior-month return (i.e., short-term reversal, which is positively priced over the same period); both channels mechanically push the high- $E[Skew]$ leg toward stocks with positive average returns, even though the pure skewness preference channel would predict the opposite sign. The fact that the skewness-managed alphas remain large and significant after controlling for $E[Skew]$ therefore indicates that the skewness-management overlay is providing genuine incremental return beyond the loadings already embedded in the standalone $E[Skew]$ portfolio.

Once we control for the original anomaly, alphas remain significant for roughly half of the strategies. In Panel B, alphas survive at the 5% level for Beta (4.43%), B/M (7.62%), OP (5.07%), investment (6.80%), Momentum (3.40%), announcement returns (6.82%), NSI (2.29%), CoEqIs12

(7.67%), and GP (5.45%). For these anomalies, the residual return after controlling for both the original anomaly portfolio and the $E[Skew]$ portfolio is economically large and statistically robust, suggesting a genuine incremental contribution of skewness management beyond simply combining the original signal with a generic skewness tilt. The exposure to the original anomaly is either indistinguishable from one or significantly below one for 17 of 18 factors (Size is the exception). Adding Fama-French five-factor controls (Panel C) leaves the qualitative picture intact. The set of anomalies with statistically significant alphas after Fama-French five-factor, original, and $E[Skew]$ controls is essentially the same set as in Panel B, with Size newly significant, and Beta, Momentum, and NSI dropping below the threshold. These results indicate that skewness management is not a redundant overlay: it generates risk-adjusted returns that, the original anomaly characteristic, the standalone expected-skewness portfolio, and the Fama-French 5-factors jointly fail to explain.

5.3 Analysis using Idiosyncratic Skewness

In this section, we replace realized skewness with idiosyncratic skewness as the target variable, then re-estimate the predictive model before re-implementing the skewness-management strategy. We construct the idiosyncratic skewness measure following Boyer et al. (2010). Specifically, we compute idiosyncratic skewness and idiosyncratic volatility for each stock-month pair from daily Fama-French three-factor model residuals. For each stock i in month t we estimate

$$r_{i,d} = \alpha_i + \beta_{i,M} \text{MKT}_d + \beta_{i,\text{SMB}} \text{SMB}_d + \beta_{i,\text{HML}} \text{HML}_d + \varepsilon_{i,d},$$

using all available trading days d within month t . Idiosyncratic volatility is the within-month standard deviation of the residuals $\hat{\varepsilon}_{i,d}$, and idiosyncratic skewness is the bias-adjusted sample skewness of those residuals. We require at least 15 daily observations per stock-month for an estimate to be retained.

$$is_{i,t} = \frac{1}{N(t)} \frac{\sum_{d=1}^{N(t)} \hat{\varepsilon}_{i,d}^3}{iv_{i,t}^3},$$

$$iv_{i,t} = \left(\frac{1}{N(t)} \sum_{d=1}^{N(t)} \epsilon_{i,d}^2 \right)^{1/2}.$$

We then repeat our predictive regressions using the idiosyncratic measure of skewness. The results are presented in Table A7. Volatility and prior-month return remain consistent predictors, though the explanatory power (adjusted R^2) is somewhat lower than with total realized skewness. This is consistent with prior findings that idiosyncratic skewness is inherently harder to forecast.

Finally, we implement the skewness-management strategy using expected idiosyncratic skewness. Table A8 shows that the results closely parallel the baseline analysis: performance gains remain substantial across nearly all anomalies. Across the 18 factors, the mean annual return improvement averages 5.35 percentage points, and Sharpe ratios rise in roughly half of the strategies. For some anomalies — such as Size and ROE — the improvements are slightly larger than under realized skewness, while for others (e.g., Beta and Investment) they are more modest. These results confirm that our findings are not specific to one measure of skewness.

5.4 Skewness-Managed Performance without Continuous Return Predictors

Finally, we examine whether the performance of the skewness-managed strategies simply reflects information already contained in the continuous return predictors used to estimate expected skewness (realized volatility, momentum, and prior month return). To test this, we re-estimate the model for expected skewness after removing these continuous return predictors, keeping only the other firm characteristics, industry, and exchange indicators.

Table A10 compares the performance of the skewness-managed anomalies in this reduced-predictor setting to the baseline. The results show that removing these continuous return predictors has little effect on the qualitative findings. Average annual return gains remain large, gains being 6.0 percentage points for model 8 and 4.7 percentage points for model 9, and Sharpe ratio improvements are similar in magnitude to the baseline. This suggests that the benefits of skewness management are not driven primarily by these standard return predictors, but by broader

relationships between firm characteristics and skewness.

Together, these robustness checks confirm that the performance improvements from skewness management generally survive transaction costs, and that they are not an artifact of the predictor set, the standalone expected skewness anomaly, or the specific definition of skewness used in our main analysis.

6 Time-Series Variation in the Skewness Premium

The analysis so far has focused on the average performance of the skewness-managed portfolios across the full sample. A natural extension is to ask whether the premium that skewness management generates above the traditional factors varies over time, and specifically whether it is related to macroeconomic conditions. To explore this, we examine a set of macroeconomic state variables including NBER recession indicators, credit spreads, cross-sectional return dispersion (CS dispersion), the VIX, market returns, and the factors from Ludvigson and Ng (2007). We examine this question using a combination of descriptive bin analyses, pooled time-series regressions, and individual factor interaction regressions that trace how the relationship between skewness management and macroeconomic conditions varies across individual anomalies.

6.1 The Equal-Weighted Increment

To examine this, we construct a single aggregate time-series: the equal-weighted average monthly increment across the 18 anomalies studied in Section 4. For each month t , we compute the return difference between the skewness-managed and standard version of each anomaly and average equally across all anomalies with non-missing data:

$$\bar{\Delta}_t = \frac{1}{N_t} \sum_{i=1}^{N_t} (R_{i,t}^{SM} - R_{i,t}^{Std}),$$

where N_t is the number of anomalies available in month t (18 for virtually the entire sample) and returns are expressed as annualized percentages. The sample spans July 1963 through December

2024 (738 months), and over this period the increment averages 5.58% per year, consistent with the cross-sectional averages reported in Table 5.

Figure 3 plots the cumulative value of a \$1 investment in the equal-weighted standard anomaly portfolio alongside the equal-weighted skewness-managed portfolio over the full sample. By December 2024, the standard portfolio grows to approximately \$22, while the skewness-managed portfolio grows to approximately \$530. The skewness-managed portfolio outperforms in 54% of sample months, with the overall cumulative advantage driven primarily by the magnitude of gains during periods of elevated outperformance rather than by consistent monthly returns. This episodic pattern suggests that the skewness premium may be related to prevailing macroeconomic conditions, a possibility we explore in the subsections that follow.

6.2 The Skewness Premium Across Macroeconomic States

Figure 4 provides a first look at how the equal-weighted increment varies with macroeconomic conditions. Panel A splits the sample into NBER expansion and recession months. The contrast is stark: the increment averages 3.7% per year during expansions but rises to 20.4% per year during recessions, a more than five-fold increase. Recession months account for only 11.5% of the sample months, yet they are associated with disproportionately large skewness management returns, suggesting that the premium is not simply a reflection of average economic conditions but concentrates during periods of economic distress.

Panel B extends the analysis to five continuously measured macroeconomic variables using quintile sorts.⁹ Four of the five variables — VIX, credit spread, CS dispersion (IQR), and market return — display a broadly similar pattern. The increment is substantially elevated in the highest

⁹We examined a broader set of candidate variables, including the term spread, the full collection of Ludvigson and Ng (2007) macro factors, and additional financial conditions measures. Several Ludvigson and Ng (2007) factors, including F4 (housing and consumption), F6 (interest rates), and F7 (stock market activity), displayed weak or difficult-to-interpret quintile patterns, and the term spread showed no consistent relationship with the increment. For brevity, we report the five variables that produce the clearest and most interpretable patterns, and carry these same variables forward through the analyses. For the real activity factor, we use the cubic transformation $F1^3$, as proposed by Ludvigson and Ng (2007) to capture nonlinear tail dynamics in real activity that the linear factor understates. This choice is supported by the regression analyses in the following sections, where $F1^3$ substantially outperforms the linear form.

quintile, with Q5 means of 18.2%, 15.1%, 16.7%, and 14.3% per year respectively, while the lower quintiles cluster near or below the unconditional mean. The real activity factor $F1^3$ of Ludvigson and Ng (2007) is the exception, displaying a U-shaped pattern in which the increment is elevated at both extremes ($Q1 = 7.8\%$, $Q5 = 11.8\%$) but near zero at the median ($Q3 = -0.3\%$).

Table A11 reports the summary statistics and pairwise correlations among the six variables. VIX, recession, credit spread, and CS dispersion — variables associated with periods of macroeconomic stress — correlate moderately with one another (0.28–0.64), confirming that while they are related, each captures a distinct dimension of financial and economic conditions. Market return is effectively orthogonal to all five other variables (all absolute correlations are less than or equal to 0.25), confirming that it represents a genuinely different source of time-series variation.

6.3 Pooled Time-Series Regressions

Table 10 reports results from pooled OLS regressions of the equal-weighted increment on each of the six macroeconomic variables individually and in two multivariate specifications. All specifications use contemporaneous rather than lagged values of the macro variables. This choice reflects our interest in understanding the economic states in which skewness management pays off, rather than in forecasting the premium out of sample. Continuous variables are standardized ($\mu = 0$, $\sigma = 1$), so coefficients are directly comparable across variables and represent the change in the annualized increment associated with a one standard deviation change.

Among the univariate specifications, $F1^3$ is the strongest predictor, with a coefficient of 9.70% per year and a t -statistic of 5.19, producing an R^2 of 0.048. VIX is the second strongest (coefficient = 12.14%, $t = 3.85$), though its sample is restricted to the post-1990 period ($N = 420$). Recession and credit spread are both statistically significant (coefficients of 16.70% and 6.67%; t -statistics of 2.08 and 2.94 respectively), while market return is marginally significant ($t = 1.66$). CS dispersion is not significant in the linear specification ($t = 1.48$), despite displaying a pronounced Q5 spike in the bin analysis of Figure 4.

Columns (7) and (8) include the variables jointly, allowing us to assess which carry independent information about the skewness premium. In Column (7), which includes all variables except VIX to preserve the full sample, F1³ dominates ($t = 4.17$) and market return retains marginal significance ($t = 1.77$), while recession, credit spread, and CS dispersion are all absorbed. Column (8) adds VIX at the cost of restricting the sample to the post-1990 period: F1³ and VIX both remain highly significant ($t = 5.86$ and $t = 2.54$), market return strengthens further ($t = 2.40$), and the R^2 reaches 0.093. Taken together, the results point to real economic activity, financial market uncertainty, and contemporaneous market returns as the three independent dimensions of time-series variation in the skewness premium.

6.4 Cross-Factor Heterogeneity

Figure 5 reports estimates from interaction regressions of the form $R_{\text{smgd},i} = \alpha_i + \beta_{1i}R_{\text{orig},i} + \beta_{2i}Z_t + \beta_{3i}(R_{\text{orig},i} \times Z_t) + \varepsilon_t$, estimated separately for each of the 18 anomalies. We use the five macro variables that produced the clearest patterns in the bin analyses and pooled regressions (recession, credit spread, F1³, VIX, and market return) for the conditioning variable Z_t . This specification can be rewritten as:

$$R_{\text{smgd},i} = \underbrace{(\alpha_i + \beta_{2i}Z_t)}_{\text{time-varying alpha}} + \underbrace{(\beta_{1i} + \beta_{3i}Z_t)}_{\text{time-varying factor loading}} \times R_{\text{orig},i} + \varepsilon_t,$$

which shows that β_2 governs how the strategy's alpha varies with macroeconomic conditions, while β_3 governs how its loading on the original factor changes across states. A positive β_2 means the strategy's alpha is higher in stressed macro environments, while a positive β_3 means the strategy amplifies its exposure to the original factor in those same states. Importantly, these two effects are independent: a strategy can earn a higher conditional alpha in bad times ($\beta_2 > 0$) while simultaneously reducing its factor exposure ($\beta_3 < 0$).

Panel A reveals that β_2 is positive and statistically significant for the vast majority of individual factor and macro variable combinations, confirming that the time-varying alpha is higher in

stressed macro states pervasively across anomalies rather than being driven by a small subset. The conditional alpha is particularly sensitive to macro conditions for 5-year Composite Equity Issuance, Announcement Return, Gross Profitability, and Book-to-Market, which display significant positive β_2 estimates across recession, credit spread, F1³, and VIX. The market return column tells a different story, with significant β_2 estimates appearing for only three anomalies. Investment-to-Assets and Gross Profitability earn more in up-market months while Size earns less, consistent with the pooled regression results showing that market return captures a distinct dimension of variation orthogonal to the stress variables. Failure Probability is a notable exception among the stress variables, as its time-varying alpha is negatively related to F1³ and VIX, suggesting that the skewness managed strategy for this anomaly actually earns less during periods of elevated real activity stress and market uncertainty.

Panel B reveals meaningful heterogeneity in the time-varying factor loading across anomalies. Market Premium (Beta) displays the most consistently positive β_3 across the stress variables — significant for recession, credit spread, F1³, and VIX, with a VIX t -statistic of 7.54 — indicating that skewness management raises the strategy’s conditional market exposure during stressed periods. Accruals shows a similarly positive pattern under VIX ($t = 5.05$). For market return, β_3 is positive and significant for Investment, Net Stock Issues, Failure Probability, and Operating Profitability, suggesting skewness management amplifies these factors’ exposure to the original return in up-market environments. By contrast, several anomalies exhibit consistently negative β_3 with respect to the stress variables, meaning their conditional factor loading falls in bad times: Momentum decouples from its original factor across all four stress variables, Investment-to-Assets shows the largest reduction in factor loading under recession ($t = -4.77$), and Return on Equity, Net Stock Issues, and ROA all display significant negative β_3 under VIX. A falling conditional factor loading is not necessarily undesirable. If the original factor underperforms in bad times, reducing exposure to it is precisely what an investor would want. Momentum crashes are the clearest example since the negative β_3 for Momentum suggests that skewness management partially decouples the strategy’s

conditional loading from momentum crashes during macro stress.

While the coefficients of primary interest are β_2 and β_3 , we also report the intercepts α_i in Appendix Figure A1. For the full-sample macro variables, the intercepts are positive across all 18 anomalies, with cross-factor means of 4.6%, 6.3%, 6.4%, and 6.9% per year for recession, credit spread, F1³, and market return respectively, confirming a meaningful baseline skewness management premium even at average macro conditions. The VIX specification is the exception, where only 13 of 18 anomalies retain a positive intercept, and only 4 of them are statistically significant. This reflects the tendency of VIX to produce larger β_2 estimates, concentrating more of each anomaly’s return in high-VIX states and leaving a smaller baseline in normal-volatility environments.

Overall, the evidence suggests that the premium of the skewness-managed portfolios relative to the original factor portfolios is strongly state dependent. The premium rises sharply during periods of economic and financial stress, including recessions, elevated volatility, and widening credit spreads. Pooled time-series regressions identify real economic activity, financial market uncertainty, and contemporaneous market returns as the primary drivers of this variation. Finally, interaction regressions show that the time-series variation in the skewness premium is pervasive across anomalies through time-varying alpha, while time-varying factor loadings play an important but more heterogeneous role.

7 Conclusion

A substantial body of research has examined skewness as both a priced source of risk and as a behavioral anomaly linked to investor preferences for lottery-like payoffs. Much less attention, however, has been devoted to how skewness interacts with the construction and performance of the traditional cross-sectional anomalies that underpin modern factor models.

We show that extreme, positively skewed stock returns play a central role in driving many well-known anomaly portfolios. The concentration of these high-skewness stocks in either the

long or short leg of an anomaly systematically influences its returns, and their presence can be predicted in advance from observable firm characteristics. Building on this insight, we introduce a simple skewness-management strategy: within each anomaly, take long positions in high-expected-skewness stocks and short positions in low-expected-skewness stocks. This transparent, easy-to-implement modification consistently improves both returns and Sharpe ratios across a broad set of anomalies. The premium also exhibits meaningful time-series variation, concentrating in periods of macroeconomic stress while generally remaining positive across anomalies even when economic conditions are at their average levels.

These gains persist after controlling for the original anomaly returns, the unconditional skewness premium, and a range of leading factor models. The fact that portfolios derived from the same underlying characteristics can still deliver large, significant alphas points to a deeper limitation: standard linear factor models do not capture the higher-moment dynamics that impact cross-sectional returns. Skewness is not merely another factor to add to the zoo, but a dimension of return behavior that interacts with existing characteristics in systematic, predictable ways.

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Table 1: **Winsorized Performance**

This table shows the performance of anomaly portfolios constructed after capping individual stock returns above the 90th, 95th, and 99th percentiles of the monthly cross-sectional return distribution. Such winsorization takes place after portfolio construction, but before performance calculations are made. The three panels show the average return premium, the Sharpe ratio, and the return skewness of each strategy. In each panel, the statistics for every sample are provided along with the difference between the original and the 90th percentile samples. The sample period is July 1963 to December 2024. The ‘t-stat’ and ‘zscore’ rows report Newey–West (12-lag) t-statistics for the mean and Memmel z-statistics for the Sharpe-ratio differences, respectively.

Sample	Beta	Size	B/M	OP	Inv	Mom	ROE	CoEqIs	AnnRet	NSI	CoEqIs12	Accrual	NOA	InvA	FailProb	OScore	GP	ROA
Return																		
original	2.29	1.44	3.35	4.21	4.93	13.90	5.96	5.74	9.36	6.40	5.19	3.69	6.37	5.57	6.05	1.43	3.36	7.68
win99	2.13	0.70	4.13	3.79	5.94	19.88	5.27	6.19	9.48	6.63	5.73	4.19	6.16	7.00	6.66	0.28	2.74	6.25
win95	-3.33	-9.91	2.60	8.70	4.28	20.03	10.88	6.41	9.50	8.13	7.39	4.09	6.37	5.68	12.33	8.31	3.67	11.94
win90	-9.57	-17.91	1.40	13.15	3.85	21.59	15.47	6.45	9.38	9.85	9.55	4.23	6.95	5.62	19.20	14.35	3.84	16.30
Difference	-11.86	-19.36	-1.95	8.94	-1.08	7.69	9.51	0.71	0.02	3.45	4.35	0.54	0.58	0.05	13.15	12.92	0.47	8.62
t-stat	-12.94	-22.17	-3.47	15.03	-2.18	5.31	13.88	1.90	0.06	8.82	8.31	1.52	2.17	0.14	14.53	22.51	1.52	10.90
Sharpe																		
Original	0.08	0.09	0.21	0.27	0.40	0.56	0.34	0.41	0.82	0.65	0.42	0.31	0.62	0.46	0.24	0.10	0.25	0.48
win99	0.08	0.04	0.26	0.24	0.47	0.81	0.30	0.44	0.83	0.67	0.45	0.34	0.59	0.55	0.26	0.02	0.20	0.39
win95	-0.13	-0.63	0.17	0.58	0.35	0.86	0.64	0.46	0.86	0.84	0.61	0.35	0.63	0.47	0.50	0.60	0.28	0.77
win90	-0.39	-1.21	0.10	0.92	0.34	0.99	0.96	0.49	0.91	1.06	0.82	0.38	0.72	0.49	0.81	1.09	0.30	1.09
Difference	-0.47	-1.30	-0.12	0.65	-0.06	0.43	0.61	0.08	0.10	0.41	0.41	0.07	0.10	0.03	0.57	0.99	0.05	0.61
zscore	-22.60	-32.29	-5.91	25.89	-2.56	14.76	23.73	5.07	3.70	17.49	15.16	2.87	5.37	1.32	23.75	26.77	3.39	21.55
Skewness																		
original	0.50	0.84	0.36	0.23	0.48	-1.42	-0.28	-0.26	0.16	-0.02	-0.03	-0.05	-0.18	0.15	-0.33	-0.21	0.20	-0.06
win99	0.41	0.87	0.44	0.15	0.62	-1.19	-0.35	-0.26	0.16	0.02	-0.07	-0.01	-0.16	0.49	-0.24	-0.24	0.13	-0.06
win95	0.29	0.79	0.38	0.30	0.52	-1.04	-0.22	-0.22	0.14	0.04	0.03	-0.09	-0.21	0.23	-0.24	-0.21	0.20	0.06
win90	0.22	0.76	0.35	0.41	0.59	-0.93	-0.12	-0.22	0.11	0.05	0.09	-0.12	-0.18	0.28	-0.23	-0.20	0.20	0.11
difference	-0.28	-0.08	-0.01	0.18	0.10	0.50	0.16	0.05	-0.06	0.06	0.13	-0.07	2.4e-3	0.12	0.10	0.01	0.01	0.18
t-stat	-2.53	-1.19	-0.23	3.49	1.37	1.91	2.38	1.04	-0.82	1.51	1.92	-1.42	0.04	2.34	2.79	0.16	0.12	2.68

Table 2: **Prediction Variable Summary Statistics**

This table reports summary statistics for the main continuous variables used in constructing expected skewness, where *rv* is realized volatility, *rs* is realized skewness, momentum is the prior-year return omitting the most recent month, *prior* is the return in the most recent month, and turnover is the monthly share turnover (sum of daily volume divided by average shares outstanding, requiring at least 15 daily observations). Each month we compute the cross-sectional mean, standard deviation, minimum, maximum, and 25th, 50th, and 75th percentiles across all sample stocks; Panel A reports the time-series average of these monthly cross-sectional statistics. Panel B reports the time-series average of the monthly cross-sectional Pearson correlations. The sample includes all common stocks trading on AMEX, Nasdaq, and NYSE from July 1963 to December 2024.

Panel A: Summary Stats							
	mean	std	min	25%	50%	75%	max
rv	0.03	0.03	8e-4	0.02	0.03	0.04	0.55
rs	0.24	1.08	-4.26	-0.28	0.20	0.75	4.48
momentum	0.14	0.60	-0.91	-0.17	0.06	0.31	11.42
prior	0.01	0.16	-0.69	-0.06	9e-4	0.07	3.03
turnover	0.12	0.39	1e-4	0.03	0.06	0.12	14.01
Panel B: Correlation Table							
	rv	rs	momentum	prior	turnover		
rv	1.00						
rs	0.20	1.00					
momentum	-0.12	-0.03	1.00				
prior	0.17	0.36	0.01	1.00			
turnover	0.29	0.07	0.14	0.15	1.00		

Table 3: **Predictive Regressions**

This table reports Fama–MacBeth (1973) estimates from monthly cross-sectional regressions of realized skewness on the predictors listed below. For each model we run a separate cross-sectional regression in every month and then average the resulting coefficients, adjusted R^2 , and number of observations over time. The row labeled *Avg.* reports the time-series average of the monthly cross-sectional coefficients; the row labeled *%Sig* reports the fraction of months in which the monthly coefficient is statistically significant at the 5% level and shares the sign of the time-series average. The sample is the universe of common stocks trading on AMEX, NYSE, and Nasdaq and is constructed so that the resulting expected skewness estimates span July 1963 through December 2024, the period used to form the skewness-managed portfolios. Momentum (mom) is the prior-year return omitting the most recent month and prior is the most recent month’s return. Dummy variables are included for NASDAQ stocks, stocks in the small (sm) and medium (med) size terciles, and Fama–French 48 industries (ind). R^2 is the time-series average of the monthly cross-sectional adjusted R^2 , and N is the average number of stocks per monthly cross-section.

Model		rv.t-1	rs.t-1	mom	turnover	nasdaq	sm	med	prior	ind	R^2	N
1	Avg.	2.80	0.01							No	0.009	4564
	%Sig	(0.77)	(0.30)									
2	Avg.	2.19	0.01	-0.02	0.14	-0.01	0.12	0.09		No	0.020	3870
	%Sig	(0.66)	(0.19)	(0.29)	(0.23)	(0.28)	(0.61)	(0.59)				
3	Avg.									yes	0.010	4518
	%Sig											
4	Avg.	1.97	0.01	-0.02	0.10	-0.02	0.12	0.08		yes	0.027	3844
	%Sig	(0.62)	(0.16)	(0.27)	(0.17)	(0.27)	(0.59)	(0.57)				
5	Avg.								-0.25	no	0.003	4564
	%Sig								(0.53)			
6	Avg.	2.25	0.02	-0.02	0.17	-0.02	0.10	0.08	-0.37	yes	0.030	3844
	%Sig	(0.69)	(0.36)	(0.26)	(0.22)	(0.29)	(0.55)	(0.55)	(0.77)			
7	Avg.	2.31	0.02	-0.02		-0.03	0.09	0.08	-0.39	yes	0.030	4139
	%Sig	(0.73)	(0.41)	(0.24)		(0.33)	(0.55)	(0.55)	(0.79)			

Table 4: **Average Returns by Portfolio**

This panel shows the average return of all 30 portfolios for 6 representative anomaly strategies. Portfolios are first sorted on the anomaly characteristic and then merged with stock level expected skewness. The second sort is done sequentially within the characteristic portfolio. The final result is 30 portfolios, based on Sequential sorts. Returns are calculated using value weighting by prior month market cap. Differences are calculated in the last column and row, while the bottom right corner provides the diagonal, long/high minus short/low.

Panel A: Skewness in Long Leg

B/M					Size					Inv				
port	high	med	low	H-L	port	high	med	low	H-L	port	high	med	low	H-L
10	23.84	16.66	11.70	12.14	10	16.65	13.24	8.80	7.84	10	23.66	17.45	11.04	12.62
9	19.25	15.23	12.45	6.80	9	14.79	13.20	9.92	4.87	9	17.57	15.20	13.67	3.91
8	17.65	14.79	10.80	6.85	8	15.45	13.49	11.74	3.71	8	16.93	13.49	11.77	5.15
7	16.73	13.68	11.18	5.55	7	16.91	13.35	10.95	5.96	7	18.25	12.47	9.80	8.44
6	13.46	13.13	10.53	2.93	6	16.15	13.87	11.56	4.59	6	15.45	12.26	11.73	3.72
5	15.13	10.94	11.03	4.10	5	14.72	13.64	11.37	3.36	5	14.62	11.00	11.05	3.57
4	12.69	11.11	10.07	2.63	4	14.52	12.55	10.90	3.62	4	14.26	12.50	11.78	2.48
3	13.43	11.61	10.07	3.37	3	14.47	13.03	11.35	3.12	3	14.35	10.77	10.53	3.81
2	12.28	12.67	12.96	-0.68	2	13.09	11.88	10.51	2.58	2	12.19	12.10	12.05	0.15
1	13.64	10.87	10.74	2.89	1	11.84	10.73	10.67	1.17	1	10.22	7.69	9.69	0.53
L-S	10.21	5.79	0.96	13.10	L-S	4.81	2.51	-1.86	5.98	L-S	13.44	9.76	1.35	13.97
	(3.23)	(2.04)	(0.37)	(3.79)		(1.37)	(0.87)	(-0.73)	(1.50)		(4.28)	(3.88)	(0.68)	(3.92)

Panel B: Skewness in Short Leg

MOM					ROE					OP				
port	high	med	low	H-L	port	high	med	low	H-L	port	high	med	low	H-L
10	19.88	17.12	17.33	2.55	10	15.31	13.07	13.26	2.05	10	16.65	14.18	11.38	5.27
9	15.29	13.17	13.60	1.69	9	13.63	12.24	12.16	1.47	9	15.52	13.42	12.50	3.01
8	15.84	13.83	11.53	4.31	8	10.19	10.30	11.67	-1.48	8	13.65	12.24	11.77	1.89
7	16.05	11.46	10.67	5.38	7	12.78	9.65	11.40	1.38	7	15.09	10.81	10.22	4.87
6	11.48	11.67	10.00	1.48	6	15.29	12.46	10.14	5.16	6	16.31	10.37	10.53	5.78
5	11.84	11.04	10.12	1.71	5	14.73	11.81	10.92	3.82	5	15.62	11.94	9.91	5.71
4	14.87	12.15	9.58	5.29	4	12.72	10.33	11.05	1.66	4	14.65	11.47	10.87	3.78
3	15.58	12.63	8.33	7.25	3	16.13	9.26	8.32	7.80	3	14.60	10.00	9.19	5.41
2	13.77	10.20	7.07	6.70	2	13.29	9.19	8.25	5.04	2	14.53	9.66	8.34	6.18
1	5.53	5.03	1.52	4.01	1	13.30	8.57	4.31	8.99	1	16.43	9.31	5.47	10.96
L-S	14.35	12.09	15.81	18.36	L-S	2.02	4.50	8.95	11.01	L-S	0.23	4.87	5.91	11.18
	(2.81)	(3.06)	(5.05)	(5.52)		(0.50)	(1.36)	(3.02)	(3.67)		(0.06)	(1.86)	(2.36)	(4.33)

Table 5: **Skewness Management Strategy Performance**

This table compares the original to the skewness managed versions of each anomaly. Panel A lists performance in the full sample in which both the characteristic and expected skewness are non-missing. Panel B lists the performance of the skewness managed version of each anomaly in which the long leg comes from the high expected skewness sample and the short leg comes from the low expected skewness sample. Panel C reports the difference. The Sharpe ratio t-statistics from Jobson and Korkie (1981) are listed in parenthesis.

	Beta	Size	B/M	OP	Inv	Mom	ROE	CoEqIs	AnnRet	NSI	CoEqIs12	Accrual	NOA	InvA	FailProb	OScore	GP	ROA
Panel A: Standard Anomaly																		
Mean	2.29	1.44	3.35	4.21	4.93	13.90	5.96	5.74	9.36	6.40	5.19	3.69	6.37	5.57	6.05	1.43	3.36	7.68
Sharpe	0.08	0.09	0.21	0.27	0.40	0.56	0.34	0.41	0.82	0.65	0.42	0.31	0.62	0.46	0.24	0.10	0.25	0.48
Skew	0.50	0.84	0.36	0.23	0.48	-1.42	-0.28	-0.26	0.16	-0.02	-0.03	-0.05	-0.18	0.15	-0.33	-0.21	0.20	-0.06
Panel B: Skewness Managed																		
Mean	8.88	5.98	13.10	11.18	13.97	18.36	11.01	9.69	16.38	10.90	13.76	8.25	7.02	11.16	8.01	5.75	10.66	10.92
Sharpe	0.24	0.22	0.54	0.60	0.57	0.68	0.55	0.38	0.76	0.76	0.85	0.32	0.33	0.46	0.30	0.31	0.51	0.57
Skew	1.71	2.04	0.72	0.21	2.17	-0.28	-0.05	4.10	0.99	0.40	0.57	1.53	0.54	1.49	-0.47	0.38	0.38	-0.11
Panel C: Difference																		
Mean	6.59	4.54	9.75	6.97	9.04	4.46	5.05	3.94	7.02	4.51	8.57	4.56	0.65	5.59	1.96	4.32	7.29	3.24
t-stat	(2.49)	(2.26)	(4.15)	(3.53)	(2.94)	(2.04)	(2.01)	(1.22)	(2.44)	(2.31)	(3.88)	(1.40)	(0.23)	(2.20)	(1.08)	(1.83)	(2.65)	(1.50)
Sharpe	0.15	0.14	0.32	0.32	0.17	0.12	0.21	-0.03	-0.06	0.11	0.44	0.01	-0.29	0.00	0.06	0.21	0.26	0.09
zscore	(2.39)	(2.04)	(2.83)	(2.91)	(1.16)	(1.45)	(1.65)	(-0.25)	(-0.38)	(0.81)	(3.34)	(0.10)	(-1.84)	(0.02)	(0.84)	(1.53)	(1.89)	(0.72)
Skew	1.21	1.20	0.36	-0.02	1.69	1.14	0.23	4.36	0.83	0.42	0.60	1.58	0.72	1.34	-0.14	0.58	0.19	-0.05
t-stat	(2.36)	(2.35)	(1.32)	(-0.06)	(2.34)	(3.69)	(0.66)	(1.88)	(1.73)	(0.93)	(1.70)	(3.35)	(2.07)	(2.69)	(-1.01)	(2.10)	(0.38)	(-0.20)

Table 6: **Skewness Managed vs Standard Anomaly Spanning Test**

This table reports alphas and betas from spanning tests of a skewness managed vs a standard long-short anomaly portfolio. The Specification is such that $R_{skew} = \alpha + \beta R_{Standard} + \epsilon$, estimated by OLS with HAC (Newey-West) standard errors using 12 monthly lags. Heteroskedasticity robust t-stats are in parenthesis. Alphas are annualized in percent. The t-statistic reported under Alpha tests the null $\alpha = 0$. The t-statistic reported under Beta tests the null $\beta = 1$, so a significantly negative value indicates $\beta < 1$ and a significantly positive value indicates $\beta > 1$. R^2 is the adjusted R^2 , RMSE is the residual standard error of the monthly regression, and N is the number of months.

	α	β	R^2	RMSE	N
Beta	6.13 (2.52)	1.20 (3.42)	0.76	0.053	738
Size	3.97 (2.19)	1.39 (4.42)	0.74	0.039	738
B/M	9.91 (4.23)	0.95 (-0.80)	0.37	0.056	738
OP	7.96 (3.97)	0.77 (-4.39)	0.39	0.042	738
Inv	10.83 (3.38)	0.64 (-2.28)	0.10	0.067	738
Mom	6.26 (2.82)	0.87 (-2.33)	0.64	0.047	738
ROE	7.16 (2.85)	0.65 (-3.94)	0.32	0.048	663
CoEqIs	4.74 (1.45)	0.86 (-1.23)	0.22	0.066	738
AnnRet	9.55 (2.85)	0.73 (-1.74)	0.15	0.058	639
NSI	6.92 (3.13)	0.62 (-3.32)	0.18	0.038	738
CoEqIs12	10.45 (4.80)	0.64 (-6.58)	0.24	0.041	738
Accrual	5.65 (1.79)	0.70 (-3.01)	0.11	0.069	738
NOA	3.95 (1.50)	0.48 (-3.69)	0.05	0.060	738
InvA	6.43 (2.51)	0.85 (-1.67)	0.18	0.063	738
FailProb	2.56 (1.53)	0.90 (-3.30)	0.73	0.040	633
OScore	4.80 (2.12)	0.66 (-4.48)	0.26	0.047	638
GP	8.47 (3.11)	0.65 (-5.30)	0.17	0.054	738
ROA	5.51 (2.32)	0.70 (-3.18)	0.35	0.045	638

Table 7: **Alphas of H-L Portfolios**

This table reports alphas from time-series factor regressions of both standard and skewness-managed long-short portfolios on six asset-pricing models: Fama-French three-factor (FF3), Fama-French five-factor (FF5), Fama-French-Carhart (FFC4), Hou-Xue-Zhang q -factor (Q4), Stambaugh-Yuan (SY4), and Daniel-Hirshleifer-Sun (DHS3). Alphas are annualized in percent. Newey-West heteroskedasticity- and autocorrelation-robust t -statistics computed with 12 monthly lags are reported in parentheses.

	α on Standard H-L Portfolio						α on Skewness Managed H-L Portfolio					
Signal	FF3	FF5	FFC4	Q4	SY4	DHS3	FF3	FF5	FFC4	Q4	SY4	DHS3
Beta	-5.42 (-2.32)	-0.86 (-0.37)	-3.00 (-1.33)	1.18 (0.42)	2.21 (0.87)	5.90 (1.84)	-0.31 (-0.09)	5.34 (1.71)	5.39 (1.33)	9.84 (2.23)	11.93 (2.62)	17.87 (2.73)
Size	-1.98 (-1.98)	-1.02 (-0.83)	0.29 (0.28)	1.62 (1.02)	1.77 (1.52)	6.90 (2.21)	0.67 (0.30)	4.82 (1.95)	4.96 (1.81)	9.48 (2.69)	8.92 (2.72)	16.24 (2.76)
B/M	-2.01 (-1.67)	-1.34 (-1.23)	-2.44 (-2.16)	-0.21 (-0.11)	-3.05 (-1.87)	3.53 (1.22)	5.22 (2.39)	6.93 (3.00)	7.32 (3.05)	9.94 (3.53)	8.19 (2.76)	17.42 (3.71)
OP	6.75 (3.36)	0.24 (0.24)	5.93 (3.28)	-0.33 (-0.20)	-1.00 (-0.60)	0.08 (0.04)	11.69 (4.65)	5.75 (2.62)	11.32 (4.43)	4.51 (1.65)	3.90 (1.55)	10.29 (3.46)
Inv	3.58 (2.64)	1.08 (1.07)	3.09 (2.39)	1.55 (1.43)	-1.83 (-1.48)	3.91 (2.00)	10.12 (3.20)	12.70 (4.01)	12.29 (4.18)	15.86 (4.22)	11.76 (3.78)	21.30 (3.94)
Mom	19.02 (7.20)	16.33 (4.35)	4.04 (3.35)	6.54 (1.63)	2.93 (1.04)	-0.15 (-0.04)	21.45 (6.95)	20.29 (5.09)	7.03 (3.20)	10.42 (2.32)	8.03 (2.60)	7.93 (2.09)
ROE	8.97 (3.89)	3.38 (2.42)	6.61 (3.43)	-0.58 (-0.37)	1.54 (0.84)	-1.89 (-0.75)	12.54 (4.18)	8.80 (3.05)	11.15 (3.91)	5.01 (1.68)	6.84 (2.42)	10.05 (3.02)
CoEqIs	8.31 (5.72)	8.20 (5.18)	5.26 (3.72)	7.12 (3.82)	6.18 (4.09)	2.86 (1.52)	8.63 (3.03)	9.25 (3.12)	10.48 (2.50)	10.92 (2.55)	10.98 (2.61)	16.32 (2.71)
AnnRet	11.24 (8.04)	10.76 (7.49)	8.67 (5.88)	9.29 (5.89)	8.10 (5.33)	0.73 (0.58)	16.80 (5.46)	18.41 (5.98)	16.45 (4.77)	18.28 (5.03)	17.59 (4.67)	14.56 (3.49)
NSI	7.76 (6.95)	4.99 (5.20)	7.19 (6.46)	5.31 (4.61)	2.22 (2.17)	2.80 (2.33)	10.64 (5.30)	8.97 (4.00)	10.09 (5.08)	8.69 (3.67)	5.01 (2.60)	9.67 (3.16)
CoEqIs12	5.41 (4.14)	3.14 (2.54)	5.30 (4.04)	4.01 (2.59)	-0.48 (-0.40)	2.34 (1.61)	12.45 (5.44)	11.15 (4.90)	13.58 (5.88)	11.81 (4.95)	8.95 (3.71)	12.50 (3.62)
Accrual	4.12 (2.77)	5.57 (4.01)	3.79 (2.71)	6.54 (4.03)	3.42 (2.58)	4.60 (2.44)	5.26 (1.50)	9.59 (2.47)	6.96 (2.14)	11.03 (2.46)	9.57 (2.59)	15.47 (3.39)
NOA	5.94 (4.12)	6.60 (4.32)	5.60 (4.39)	5.60 (3.40)	3.63 (3.18)	4.45 (3.31)	5.39 (2.07)	10.35 (3.91)	6.70 (2.55)	11.34 (3.62)	8.63 (3.06)	15.20 (3.83)
InvA	4.41 (3.22)	3.06 (2.47)	3.52 (2.63)	2.26 (1.59)	-0.68 (-0.45)	4.02 (2.16)	6.23 (2.27)	6.46 (2.31)	7.98 (2.66)	8.06 (2.38)	4.66 (1.41)	12.55 (2.67)
FailProb	12.28 (4.29)	6.87 (2.21)	3.87 (1.55)	2.19 (0.75)	-3.20 (-1.58)	-7.39 (-2.32)	14.31 (4.69)	10.69 (2.77)	5.04 (2.20)	4.97 (1.35)	-1.52 (-0.55)	-3.26 (-0.93)
OScore	4.40 (2.74)	3.60 (2.06)	4.30 (2.79)	2.57 (1.48)	2.12 (1.32)	-0.91 (-0.40)	6.49 (2.10)	7.24 (1.89)	7.15 (2.42)	6.66 (1.77)	6.10 (1.67)	7.53 (2.32)
GP	7.91 (5.83)	4.10 (3.19)	6.22 (4.66)	4.22 (2.32)	2.57 (1.97)	2.17 (1.08)	12.54 (5.30)	11.99 (4.47)	12.56 (4.87)	13.54 (3.58)	10.32 (3.46)	15.15 (3.76)
ROA	10.90 (5.10)	5.71 (4.50)	7.86 (3.97)	1.48 (1.19)	2.61 (1.81)	0.75 (0.33)	13.12 (4.39)	9.56 (3.16)	10.74 (3.41)	5.70 (1.79)	5.83 (1.81)	8.04 (2.29)

Table 8: **Changes in Univariate Betas due to Skewness Management**

This table reports the difference in univariate factor betas between the skewness-managed and the standard long-short anomaly portfolios. For each anomaly the difference series $r_t^{managed} - r_t^{standard}$ is regressed on each individual asset-pricing factor in a separate univariate regression: $r_t^{diff} = a + \beta_f f_{j,t} + \varepsilon_t$. The reported coefficient is therefore $\beta_f^{managed} - \beta_f^{standard}$, the change in univariate factor exposure induced by the skewness tilt. Newey-West heteroskedasticity- and autocorrelation-consistent t-statistics with 12 monthly lags are reported in parentheses.

	MKT	SMB	HML	UMD	RMW	CMA	ME	IA	ROE	MGMT	PERF	PEAD	FIN
Beta	0.25 (4.50)	0.43 (4.15)	-0.19 (-1.61)	-0.34 (-2.20)	-0.54 (-5.12)	-0.23 (-1.60)	0.41 (4.01)	-0.25 (-1.44)	-0.65 (-3.29)	-0.37 (-4.10)	-0.38 (-3.75)	-0.40 (-1.43)	-0.35 (-4.68)
Size	0.37 (6.80)	0.50 (4.13)	-0.33 (-1.92)	-0.23 (-1.50)	-0.95 (-5.28)	-0.47 (-2.26)	0.47 (3.68)	-0.55 (-2.16)	-0.80 (-4.13)	-0.58 (-4.06)	-0.33 (-3.15)	-0.08 (-0.29)	-0.63 (-5.22)
B/M	0.37 (5.74)	0.71 (7.11)	-0.22 (-1.39)	-0.29 (-2.02)	-0.70 (-4.68)	-0.31 (-1.61)	0.68 (6.81)	-0.33 (-1.40)	-0.74 (-4.66)	-0.53 (-4.37)	-0.38 (-3.55)	-0.14 (-0.52)	-0.58 (-6.50)
OP	0.23 (4.37)	0.46 (5.45)	-0.03 (-0.26)	-0.10 (-0.85)	-0.40 (-3.25)	-0.11 (-0.94)	0.44 (4.86)	-0.06 (-0.37)	-0.34 (-2.89)	-0.21 (-2.58)	-0.13 (-1.61)	-0.07 (-0.27)	-0.28 (-3.23)
Inv	0.47 (6.58)	1.02 (4.83)	-0.45 (-1.56)	-0.30 (-1.24)	-1.47 (-4.30)	-0.73 (-2.68)	0.98 (4.20)	-0.92 (-2.49)	-1.19 (-4.51)	-0.90 (-4.49)	-0.49 (-3.23)	0.02 (0.04)	-0.93 (-3.81)
Mom	0.23 (5.00)	0.54 (5.68)	-0.05 (-0.34)	-0.10 (-1.02)	-0.66 (-3.87)	-0.10 (-0.59)	0.51 (5.16)	-0.01 (-0.06)	-0.51 (-3.65)	-0.31 (-2.82)	-0.27 (-4.15)	-0.10 (-0.50)	-0.36 (-3.36)
ROE	0.27 (3.81)	0.53 (4.70)	-0.16 (-0.91)	-0.15 (-1.15)	-0.75 (-4.49)	-0.21 (-0.90)	0.51 (4.56)	-0.21 (-0.86)	-0.57 (-3.72)	-0.36 (-2.27)	-0.24 (-2.39)	-0.19 (-0.63)	-0.46 (-3.36)
CoEqIs	0.38 (4.96)	0.73 (5.44)	-3e-3 (-0.02)	-0.55 (-2.47)	-0.54 (-3.25)	-0.21 (-1.11)	0.70 (5.03)	-0.17 (-0.72)	-0.87 (-3.13)	-0.39 (-3.51)	-0.49 (-3.13)	-0.76 (-1.52)	-0.46 (-4.69)
AnnRet	0.27 (4.30)	0.73 (4.10)	-0.22 (-1.13)	-0.25 (-1.41)	-0.94 (-3.69)	-0.22 (-1.08)	0.70 (3.88)	-0.28 (-1.16)	-0.80 (-4.31)	-0.49 (-3.69)	-0.35 (-3.52)	-0.25 (-0.66)	-0.52 (-3.32)
NSI	0.23 (5.19)	0.51 (3.78)	-0.14 (-0.88)	-0.04 (-0.29)	-0.56 (-2.30)	-0.19 (-1.44)	0.50 (3.37)	-0.19 (-0.98)	-0.40 (-2.78)	-0.31 (-2.63)	-0.10 (-1.18)	0.13 (0.40)	-0.41 (-2.68)
CoEqIs12	0.23 (4.84)	0.37 (4.88)	-0.14 (-1.27)	-0.15 (-1.52)	-0.48 (-4.41)	-0.19 (-1.31)	0.36 (4.11)	-0.15 (-0.87)	-0.42 (-3.83)	-0.32 (-3.69)	-0.22 (-2.83)	-0.12 (-0.59)	-0.36 (-5.36)
Accrual	0.42 (5.03)	0.97 (5.73)	-0.16 (-0.60)	-0.27 (-1.45)	-1.19 (-3.74)	-0.27 (-0.88)	0.94 (5.11)	-0.28 (-0.71)	-0.95 (-3.89)	-0.61 (-2.89)	-0.49 (-3.30)	-0.08 (-0.17)	-0.77 (-4.15)
NOA	0.41 (4.88)	0.85 (4.88)	-0.61 (-2.44)	-0.16 (-0.79)	-1.42 (-5.63)	-0.68 (-2.37)	0.80 (4.54)	-0.85 (-2.69)	-1.03 (-4.29)	-0.89 (-4.59)	-0.35 (-2.65)	0.01 (0.03)	-0.99 (-5.86)
InvA	0.42 (6.24)	0.81 (7.30)	-0.06 (-0.35)	-0.35 (-2.18)	-0.79 (-4.45)	-0.24 (-1.52)	0.80 (7.28)	-0.25 (-1.14)	-0.89 (-5.11)	-0.46 (-5.51)	-0.44 (-3.48)	-0.24 (-0.66)	-0.51 (-5.08)
FailProb	0.11 (3.07)	0.37 (1.66)	-0.24 (-1.45)	0.08 (0.69)	-0.66 (-2.38)	-0.19 (-1.76)	0.35 (1.62)	-0.27 (-1.74)	-0.25 (-1.33)	-0.24 (-2.52)	-0.03 (-0.45)	0.26 (1.07)	-0.39 (-2.59)
OScore	0.29 (5.07)	0.61 (4.86)	-0.08 (-0.41)	-0.14 (-1.07)	-0.69 (-2.97)	-0.17 (-0.77)	0.58 (4.64)	-0.21 (-0.84)	-0.55 (-3.62)	-0.37 (-2.24)	-0.24 (-2.70)	0.09 (0.32)	-0.48 (-3.46)
GP	0.38 (5.59)	0.66 (4.63)	-0.21 (-0.88)	-0.22 (-1.24)	-1.05 (-4.31)	-0.34 (-1.32)	0.63 (4.42)	-0.33 (-1.07)	-0.90 (-3.81)	-0.46 (-2.53)	-0.37 (-2.81)	-0.16 (-0.42)	-0.68 (-4.22)
ROA	0.17 (2.94)	0.44 (2.81)	-0.12 (-0.63)	-0.09 (-0.72)	-0.66 (-2.87)	-0.09 (-0.41)	0.41 (2.65)	-0.17 (-0.65)	-0.45 (-2.68)	-0.25 (-1.51)	-0.16 (-1.67)	0.02 (0.07)	-0.40 (-2.45)

Table 9: **Changes in Multivariate Betas due to Skewness Management**

This table contains differences in β from multi-factor regressions on skewness-managed and standard anomaly portfolios. For each anomaly the difference series $r_t^{managed} - r_t^{standard}$ is regressed on each asset-pricing model's factor block; the reported coefficient is therefore $\beta_f^{managed} - \beta_f^{standard}$. Newey-West HAC t-statistics with 12 monthly lags are reported in parentheses.

	FF3			DHS3			FFC4				SY4			
	MKT	SMB	HML	MKT	PEAD	FIN	MKT	SMB	HML	UMD	MKT	SMB	MGMT	PERF
Beta	0.16 (2.64)	0.37 (3.63)	-0.15 (-1.41)	0.09 (1.19)	-0.40 (-1.55)	-0.30 (-2.66)	0.09 (1.92)	0.36 (2.98)	-0.26 (-2.41)	-0.33 (-2.41)	0.04 (0.72)	0.26 (2.36)	-0.23 (-2.36)	-0.32 (-3.42)
Size	0.26 (4.87)	0.40 (3.21)	-0.25 (-1.80)	0.16 (2.53)	-0.09 (-0.54)	-0.54 (-3.44)	0.22 (5.18)	0.39 (2.79)	-0.32 (-2.13)	-0.20 (-1.77)	0.14 (2.29)	0.28 (2.00)	-0.38 (-2.77)	-0.23 (-2.56)
B/M	0.23 (3.66)	0.61 (6.45)	-0.15 (-1.32)	0.10 (1.60)	-0.16 (-0.88)	-0.52 (-4.46)	0.18 (3.28)	0.61 (5.60)	-0.24 (-1.95)	-0.26 (-2.41)	0.13 (1.74)	0.51 (4.99)	-0.25 (-1.74)	-0.24 (-3.00)
OP	0.15 (2.95)	0.39 (4.61)	0.01 (0.15)	0.17 (2.27)	-0.05 (-0.22)	-0.18 (-1.53)	0.14 (3.10)	0.39 (4.40)	-0.00 (-0.02)	-0.05 (-0.54)	0.16 (3.25)	0.40 (4.48)	0.03 (0.42)	-0.02 (-0.26)
Inv	0.24 (3.60)	0.92 (4.42)	-0.39 (-1.71)	0.08 (0.74)	-0.03 (-0.09)	-0.88 (-2.99)	0.19 (4.13)	0.91 (3.93)	-0.48 (-2.15)	-0.27 (-1.62)	0.05 (0.68)	0.73 (3.09)	-0.62 (-3.57)	-0.33 (-2.48)
Mom	0.14 (2.97)	0.48 (4.54)	-0.01 (-0.12)	0.11 (1.65)	-0.10 (-0.72)	-0.30 (-2.17)	0.12 (3.20)	0.48 (4.35)	-0.03 (-0.29)	-0.06 (-0.82)	0.07 (1.47)	0.43 (4.02)	-0.11 (-0.95)	-0.17 (-3.11)
ROE	0.17 (2.86)	0.46 (4.25)	-0.11 (-0.80)	0.11 (1.50)	-0.20 (-0.93)	-0.40 (-2.41)	0.15 (2.81)	0.46 (3.83)	-0.14 (-1.06)	-0.10 (-1.05)	0.13 (2.11)	0.41 (3.38)	-0.13 (-0.81)	-0.10 (-1.07)
CoEqIs	0.27 (3.75)	0.62 (5.03)	0.08 (0.54)	0.18 (1.83)	-0.74 (-1.73)	-0.37 (-2.31)	0.17 (2.64)	0.61 (4.23)	-0.09 (-0.65)	-0.50 (-2.51)	0.17 (2.24)	0.54 (4.83)	-0.07 (-0.58)	-0.34 (-2.51)
AnnRet	0.13 (1.89)	0.68 (3.72)	-0.19 (-1.30)	0.03 (0.28)	-0.28 (-1.05)	-0.51 (-2.46)	0.08 (1.51)	0.67 (3.25)	-0.27 (-1.82)	-0.22 (-1.67)	0.03 (0.46)	0.57 (2.71)	-0.28 (-2.30)	-0.22 (-2.17)
NSI	0.13 (2.69)	0.46 (3.20)	-0.11 (-0.82)	0.11 (1.34)	0.13 (0.56)	-0.34 (-1.83)	0.13 (3.41)	0.46 (3.16)	-0.11 (-0.97)	-0.00 (-0.02)	0.13 (3.02)	0.44 (3.08)	-0.07 (-0.75)	0.02 (0.32)
CoEqIs12	0.16 (2.97)	0.31 (3.70)	-0.10 (-1.07)	0.15 (2.63)	-0.10 (-0.70)	-0.28 (-3.11)	0.14 (3.01)	0.31 (3.36)	-0.14 (-1.57)	-0.13 (-1.66)	0.10 (1.90)	0.25 (2.53)	-0.16 (-1.59)	-0.14 (-2.00)
Accrual	0.25 (3.08)	0.86 (4.86)	-0.10 (-0.44)	0.07 (0.72)	-0.12 (-0.41)	-0.73 (-3.12)	0.21 (3.05)	0.86 (4.40)	-0.16 (-0.73)	-0.21 (-1.53)	0.12 (1.38)	0.75 (3.90)	-0.26 (-1.10)	-0.31 (-2.40)
NOA	0.19 (2.76)	0.78 (5.00)	-0.56 (-2.84)	0.07 (0.91)	-0.05 (-0.32)	-0.95 (-4.66)	0.15 (2.96)	0.78 (4.54)	-0.62 (-3.13)	-0.17 (-1.29)	0.03 (0.40)	0.59 (3.23)	-0.67 (-3.75)	-0.21 (-2.11)
InvA	0.29 (4.28)	0.69 (5.92)	0.03 (0.23)	0.18 (2.34)	-0.23 (-0.85)	-0.41 (-3.08)	0.24 (3.37)	0.69 (5.16)	-0.06 (-0.51)	-0.27 (-2.19)	0.21 (2.59)	0.63 (5.17)	-0.08 (-0.79)	-0.26 (-2.52)
FailProb	0.02 (0.27)	0.36 (1.63)	-0.24 (-1.55)	-0.01 (-0.15)	0.22 (1.49)	-0.39 (-2.17)	0.04 (0.63)	0.37 (1.69)	-0.21 (-1.61)	0.09 (1.03)	0.03 (0.53)	0.34 (1.44)	-0.14 (-1.66)	0.05 (0.66)
OScore	0.19 (3.51)	0.53 (3.87)	-0.04 (-0.23)	0.08 (1.21)	0.07 (0.39)	-0.43 (-2.60)	0.18 (3.74)	0.53 (3.60)	-0.06 (-0.39)	-0.08 (-0.82)	0.14 (2.18)	0.49 (3.43)	-0.12 (-0.69)	-0.10 (-1.05)
GP	0.26 (4.28)	0.55 (3.80)	-0.13 (-0.70)	0.17 (2.01)	-0.17 (-0.65)	-0.58 (-2.86)	0.23 (5.33)	0.55 (3.42)	-0.19 (-0.99)	-0.17 (-1.22)	0.18 (2.93)	0.47 (3.04)	-0.16 (-0.86)	-0.23 (-1.72)
ROA	0.09 (1.69)	0.41 (2.54)	-0.10 (-0.61)	0.01 (0.13)	-0.01 (-0.04)	-0.39 (-2.06)	0.08 (1.57)	0.40 (2.39)	-0.12 (-0.74)	-0.07 (-0.62)	0.06 (0.95)	0.37 (2.22)	-0.09 (-0.55)	-0.06 (-0.56)

Table 9: **Changes in Multivariate Betas due to Skewness Management (Cont'd)**

This table contains differences in β from multi-factor regressions on skewness-managed and standard anomaly portfolios. For each anomaly the difference series $r_t^{managed} - r_t^{standard}$ is regressed on each asset-pricing model's factor block; the reported coefficient is therefore $\beta_f^{managed} - \beta_f^{standard}$. Newey-West HAC t-statistics with 12 monthly lags are reported in parentheses.

	Q4				FF5				
	MKT	ME	IA	ROE	MKT	SMB	HML	RMW	CMA
Beta	0.14 (2.33)	0.21 (2.49)	-0.08 (-0.54)	-0.51 (-2.79)	0.15 (2.40)	0.29 (3.27)	-0.17 (-1.65)	-0.32 (-3.59)	0.09 (0.61)
Size	0.21 (4.07)	0.20 (1.77)	-0.32 (-1.74)	-0.63 (-4.01)	0.22 (4.96)	0.22 (3.56)	-0.17 (-2.03)	-0.74 (-6.35)	-0.11 (-0.83)
B/M	0.22 (3.97)	0.46 (4.75)	-0.06 (-0.34)	-0.48 (-3.29)	0.23 (3.45)	0.54 (6.14)	-0.20 (-1.78)	-0.33 (-2.86)	0.13 (0.72)
OP	0.16 (3.13)	0.34 (3.91)	0.12 (0.96)	-0.15 (-1.64)	0.15 (3.07)	0.35 (5.03)	0.00 (0.00)	-0.17 (-1.72)	0.05 (0.35)
Inv	0.14 (1.91)	0.64 (3.26)	-0.67 (-2.49)	-0.87 (-4.61)	0.16 (2.68)	0.65 (6.35)	-0.18 (-1.07)	-1.08 (-4.98)	-0.35 (-1.97)
Mom	0.16 (3.34)	0.37 (3.90)	0.18 (1.24)	-0.32 (-2.54)	0.13 (2.83)	0.38 (5.18)	-0.03 (-0.24)	-0.42 (-2.82)	0.07 (0.52)
ROE	0.15 (3.06)	0.34 (3.19)	-0.02 (-0.12)	-0.37 (-2.73)	0.16 (3.27)	0.33 (3.87)	-0.10 (-1.05)	-0.51 (-3.41)	0.08 (0.43)
CoEqIs	0.25 (3.85)	0.44 (4.80)	0.13 (0.69)	-0.61 (-2.40)	0.26 (3.51)	0.58 (4.12)	0.10 (0.63)	-0.17 (-0.93)	-0.03 (-0.12)
AnnRet	0.10 (1.48)	0.49 (3.00)	-0.11 (-0.65)	-0.57 (-4.00)	0.12 (1.81)	0.52 (4.71)	-0.20 (-1.68)	-0.60 (-3.38)	0.15 (0.89)
NSI	0.13 (2.60)	0.39 (2.69)	-0.02 (-0.13)	-0.20 (-2.05)	0.13 (2.83)	0.38 (4.38)	-0.12 (-1.09)	-0.31 (-1.81)	0.07 (0.68)
CoEqIs12	0.17 (3.88)	0.21 (2.29)	0.03 (0.21)	-0.27 (-2.81)	0.16 (3.48)	0.24 (3.30)	-0.12 (-1.32)	-0.29 (-3.12)	0.08 (0.57)
Accrual	0.24 (3.00)	0.68 (3.92)	0.03 (0.10)	-0.61 (-2.93)	0.22 (2.71)	0.68 (5.80)	-0.09 (-0.57)	-0.76 (-2.82)	0.07 (0.27)
NOA	0.14 (2.26)	0.49 (3.50)	-0.63 (-2.86)	-0.77 (-4.49)	0.14 (3.58)	0.53 (6.64)	-0.50 (-4.36)	-1.06 (-8.28)	-0.03 (-0.15)
InvA	0.27 (3.42)	0.53 (6.38)	0.07 (0.39)	-0.59 (-3.92)	0.28 (3.52)	0.60 (6.01)	0.04 (0.37)	-0.40 (-2.52)	0.01 (0.04)
FailProb	0.01 (0.19)	0.30 (1.49)	-0.22 (-1.50)	-0.12 (-1.01)	0.01 (0.13)	0.23 (1.64)	-0.24 (-1.63)	-0.51 (-2.90)	0.12 (0.91)
OScore	0.17 (3.26)	0.42 (3.43)	-0.02 (-0.09)	-0.32 (-2.63)	0.18 (3.26)	0.42 (5.06)	-0.01 (-0.07)	-0.40 (-2.07)	0.03 (0.14)
GP	0.23 (4.03)	0.35 (2.83)	-0.07 (-0.28)	-0.68 (-3.18)	0.22 (3.98)	0.36 (4.15)	-0.07 (-0.57)	-0.78 (-3.83)	-0.06 (-0.34)
ROA	0.07 (1.34)	0.29 (2.00)	-0.06 (-0.26)	-0.31 (-2.18)	0.08 (1.55)	0.27 (3.03)	-0.11 (-1.09)	-0.48 (-2.70)	0.13 (0.74)

Table 10: **Pooled Time-Series Regressions of the Equal-Weighted Increment on Macroeconomic Variables**

The table reports the coefficient estimates from OLS regressions of the equal-weighted average increment across 18 anomalies (annualized, % per year) on contemporaneous macroeconomic variables, with Newey-West HAC standard errors (12 lags). Continuous variables are standardized (mean 0, SD 1). Coefficients represent the change in the annualized increment associated with a one standard deviation increase. Recession is an NBER binary indicator (0/1). Specification (7) includes all variables except VIX to preserve the full sample ($N = 737$); specification (8) adds VIX and is restricted to the post-1990 sample ($N = 420$). t -statistics are shown in parentheses.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Recession	Credit Spread	CS Disp (IQR)	VIX	Market Return	F1 ³	Multi-variate	Multi +VIX
Constant	3.65** (2.19)	5.32*** (3.07)	5.26*** (3.08)	2.02 (0.86)	5.57*** (3.08)	5.56*** (3.26)	5.30*** (3.16)	2.33 (1.16)
Recession	16.70** (2.08)						0.73 (0.09)	-1.32 (-0.10)
Credit Spread		6.67*** (2.94)					2.50 (1.23)	-1.46 (-0.31)
CS Disp (IQR)			4.66 (1.48)				1.01 (0.36)	-2.80 (-0.73)
VIX				12.14*** (3.85)				12.27** (2.54)
Market Return					3.15* (1.66)		3.60* (1.77)	8.40** (2.40)
F1 ³						9.70*** (5.19)	8.54*** (4.17)	9.39*** (5.86)
R^2	0.014	0.022	0.010	0.050	0.005	0.048	0.059	0.093
Adj. R^2	0.012	0.021	0.009	0.048	0.003	0.046	0.052	0.080
N	737	737	737	420	737	737	737	420

Figure 1: Standard vs. Skewness Managed Anomaly Alphas The figure below plots annualized alphas from factor-model time-series regressions of standard and skewness-managed long-short portfolios. Each anomaly is evaluated against the factor model for which it is a constituent (or, for Market Premium, the average across all four models). The blue dot is the alpha of the standard long-short portfolio and the orange dot is the alpha of the skewness-managed portfolio. The connecting line indicates the direction and magnitude of the improvement.

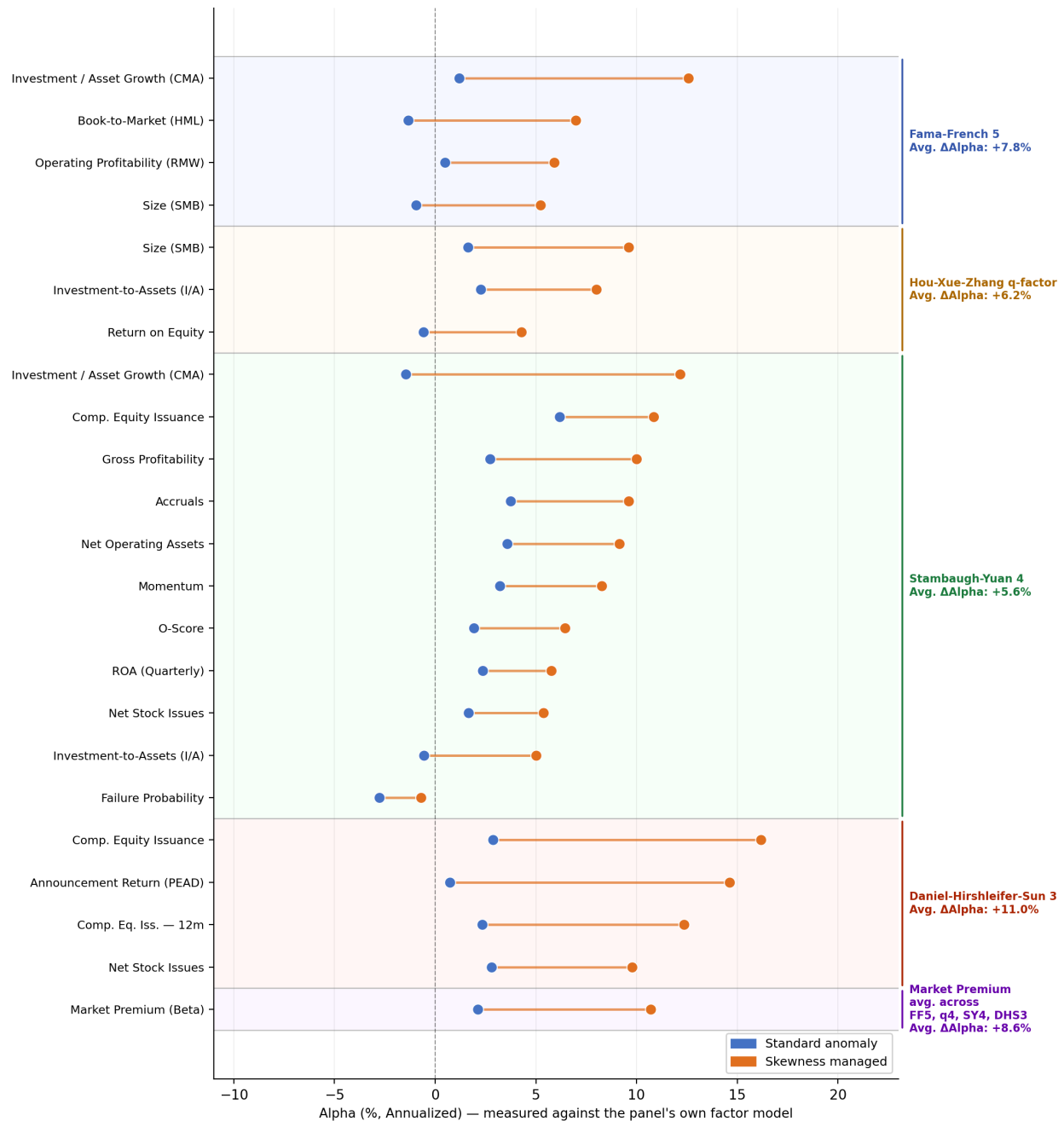


Figure 2: Characterizing Skewness in Monthly Returns

Panel A plots the empirical distribution of monthly stock returns from June 1963–December 2024 ($N = 3,416,924$). The mean return is 1.1%, the median is 0.0%, and the skewness is 9.06. The dashed curve is a fitted normal distribution ($\mu = 1.1\%$, $\sigma = 10.3\%$). The right panel magnifies the extreme positive tail (returns above 100%) on a log scale, containing 9,978 observations (0.29% of the sample). The normal distribution predicts fewer than 0.001 observations in this range. Panel B plots the percentage of stocks in a given month with returns above 26.8%, the 95th percentile of the full-sample distribution. The red line is a 12-month rolling average and the dashed orange line marks the expected 5% level. Gray shading indicates NBER recession periods. The three labeled episodes highlight elevated extreme returns following the Dot-com recession, the Financial Crisis, and the COVID-19 crash.

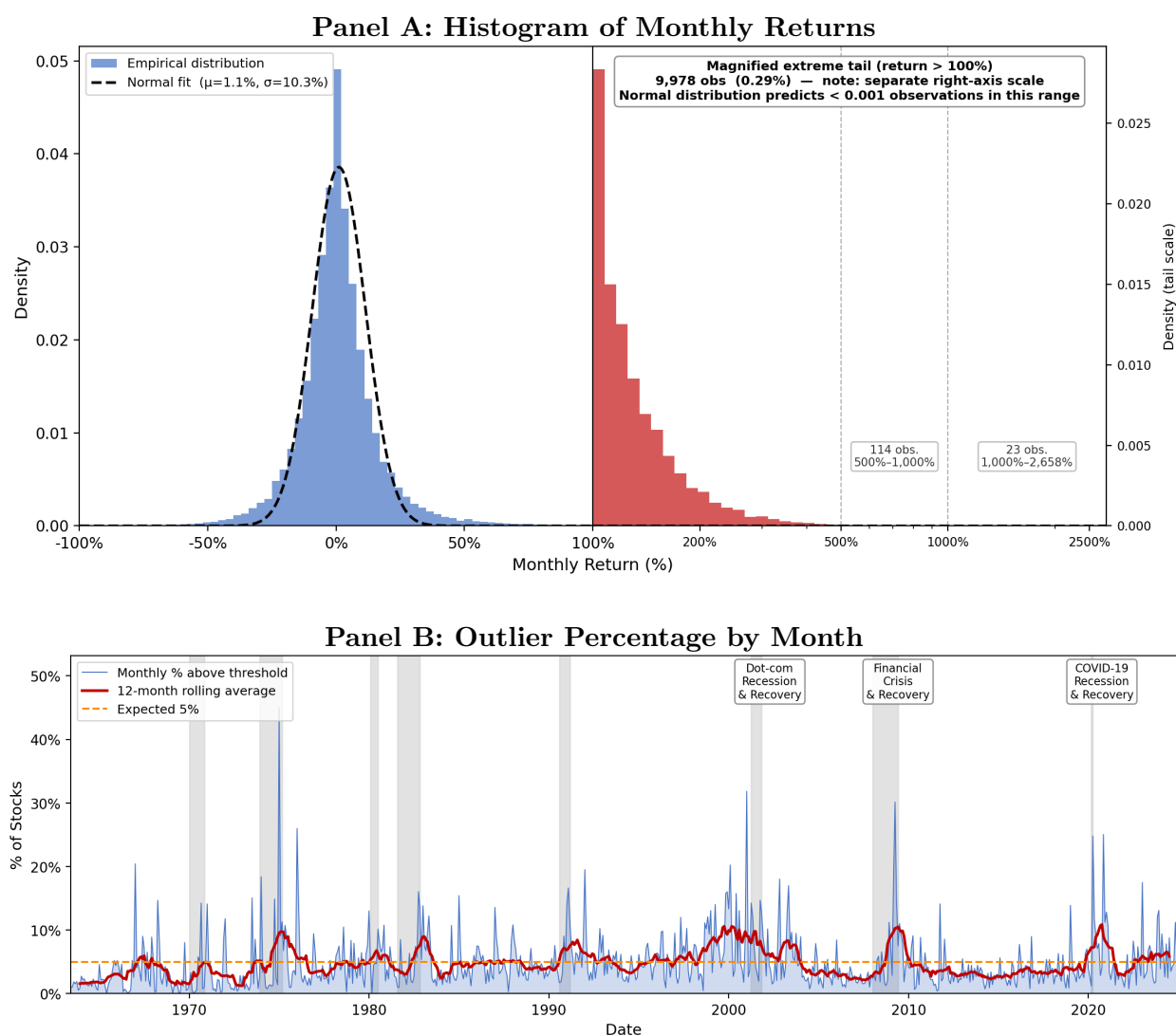


Figure 3: Cumulative Returns: Standard vs. Skewness-Managed Anomaly Portfolios

The figure plots the cumulative value of a \$1 investment in the equal-weighted standard anomaly portfolio (black) and the equal-weighted skewness-managed portfolio (blue) over the period July 1963 through December 2024. Each month, returns are averaged equally across the 18 anomalies with non-missing data. The skewness-managed portfolio applies the double-sort procedure described in Section 4, selecting the high expected-skewness tercile in the long leg and the low expected-skewness tercile in the short leg of each anomaly. The vertical axis uses a log scale.

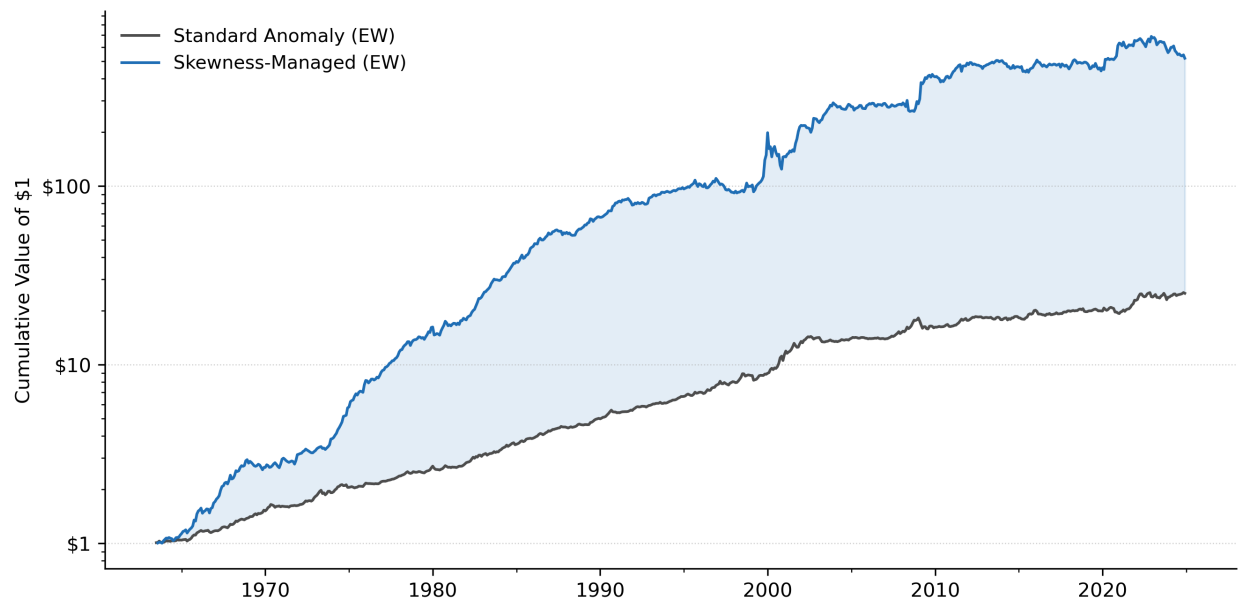


Figure 4: The Skewness Premium Across Macroeconomic States Panel A shows the mean annualized equal-weighted increment in NBER expansion and recession months over the period July 1963 through December 2024. Error bars indicate ± 1 standard error. Panel B plots the mean annualized increment across quintiles of five macroeconomic variables: credit spread (Baa–Aaa), cross-sectional return dispersion (IQR), VIX, the nonlinear real activity factor $F1^3$ of Ludvigson and Ng (2007), and contemporaneous market return. Q1 denotes the lowest quintile and Q5 the highest. VIX quintiles are formed over the post-1990 sample. All other variables use the full sample. The dashed line indicates the unconditional mean increment of 5.6% per year.

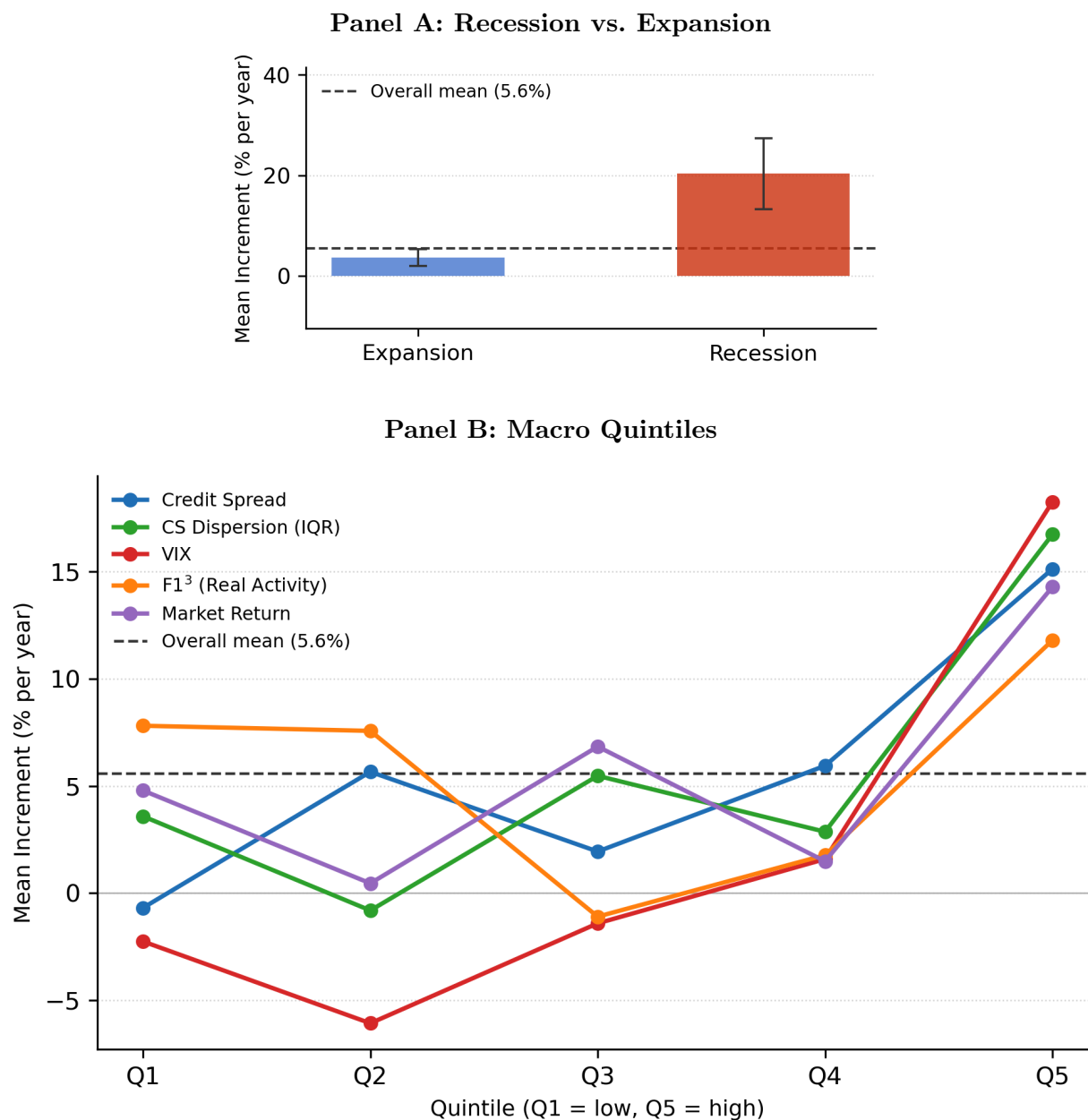
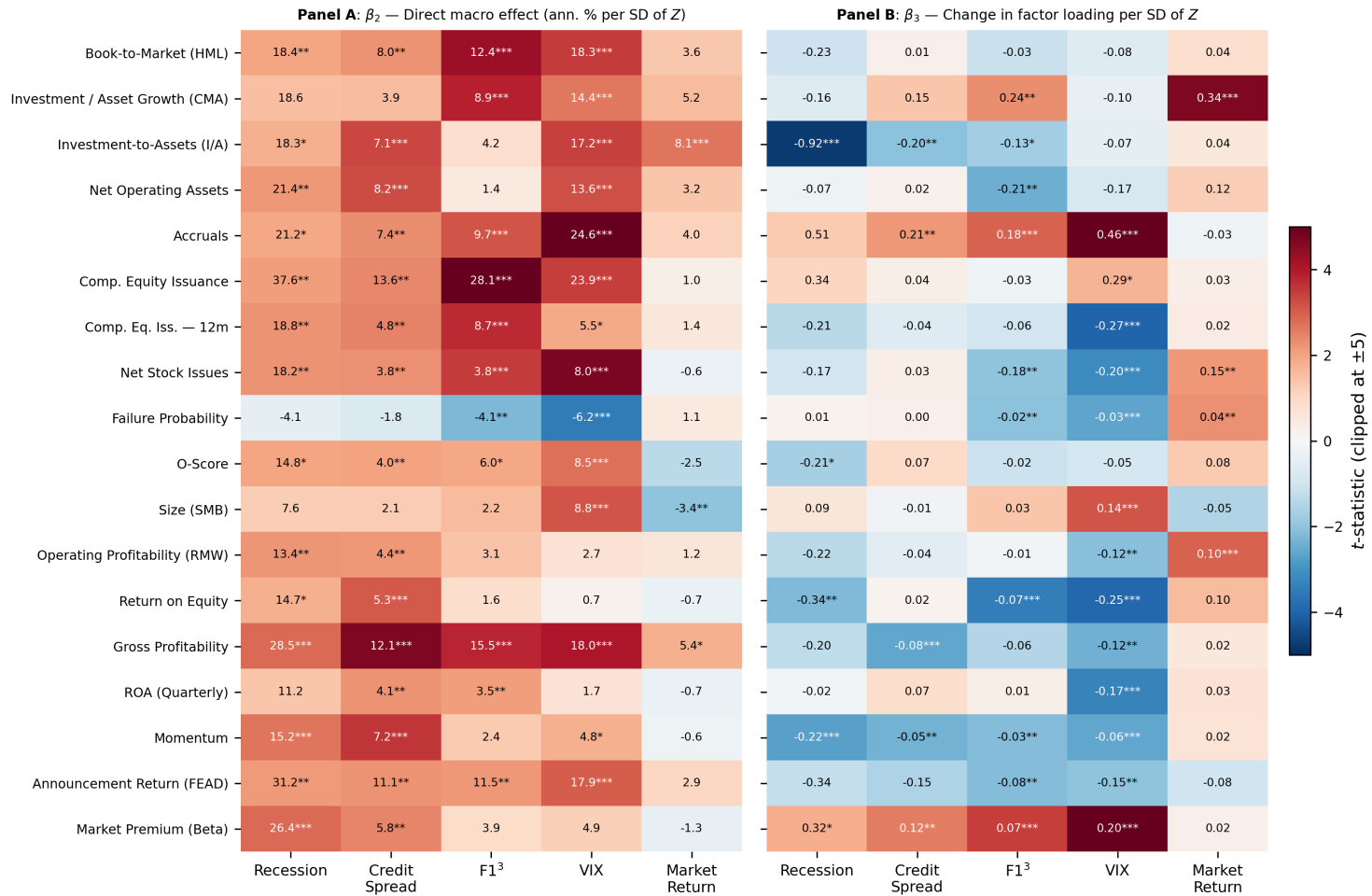


Figure 5: Factor Interaction Regressions The figure reports estimates from individual factor interaction regressions of the form $R_{\text{smgd},i} = \alpha_i + \beta_{1i}R_{\text{orig},i} + \beta_{2i}Z_t + \beta_{3i}(R_{\text{orig},i} \times Z_t) + \varepsilon_t$, estimated separately for each of the 18 anomalies. Panel A reports β_2 , the direct effect of the macroeconomic variable on the skewness-managed return (annualized, % per year per standard deviation of Z). Panel B reports β_3 , the change in the skewness-managed portfolio's loading on the original factor per standard deviation of Z . Cell values show the estimated coefficient with significance stars. Colors reflect the t -statistic on a scale clipped at ± 5 . The macro variables Z are NBER recession (0/1), credit spread, F1³, VIX, and the excess market return; continuous variables are standardized to mean zero and unit standard deviation. VIX specifications use the post-1990 sample, and all others span July 1963 through December 2024. Newey-West HAC standard errors with 12 lags.



Appendices

Anomaly Construction

This appendix details the construction of the eighteen anomalies used throughout the paper. Each signal is implemented as a value-weighted long-short decile portfolio. *Portfolios are rebalanced monthly.* At the end of each month, decile breakpoints are recomputed from the cross-section of NYSE common stocks with a non-missing signal and a positive market capitalization, and all common stocks (CRSP share codes 10 and 11) on the NYSE/AMEX/NASDAQ are then assigned to deciles based on those breakpoints. The long-short portfolio is the value-weighted return on decile 10 minus decile 1, formed using market capitalization at month end. The full sample runs from July 1963 to December 2024; strategies that rely on quarterly Compustat data (ROE and ROA) begin in January 1972, when broad quarterly coverage becomes available. Months with fewer than 200 stocks are dropped. Returns are adjusted for delisting.

For nine of the eighteen signals (Size, NSI, InvA, Inv, CoEqIs12, FailProb, Accrual, OScore, NOA), the raw characteristic is negatively related to expected returns, so the decile assignment is sign-flipped, so that decile 10 is the high-expected-return leg in every case. The remaining nine signals (and all four from Chen and Zimmermann (2022)) are constructed so that decile 10 is already the high-expected-return leg.

Signals constructed from CRSP and Compustat

- **Size (Size).** Market capitalization $|prc| \times shrout$, measured at the end of each month from the CRSP monthly file. The decile assignment is recomputed each month using monthly NYSE breakpoints and is sign-flipped because small firms earn higher expected returns.
- **Momentum (Mom).** Cumulative return from month $t - 12$ through month $t - 2$, skipping the most recent month, following Jegadeesh and Titman (1993). Computed each month from CRSP monthly returns. Stocks with a missing return in any of the eleven required months are dropped.
- **Operating Profitability (OP).** Annual revenues less cost of goods sold, interest expense, and selling, general, and administrative expenses, scaled by book equity for the most recent fiscal year ending in $t - 1$, $(REVT - COGS - XSGA - XINT)/BE$, following Fama and French (2015). Missing components of the numerator are treated as zero; the numerator must contain at least one non-missing element. The annual signal is dated four months after `datadate` and forward-filled monthly.
- **Investment (Inv).** Annual asset growth, AT_t/AT_{t-1} , following Hou et al. (2015). Forward-filled monthly with the four-month reporting lag. Sign- flipped.

- **Investment-to-Assets (InvA).** Change in gross plant plus change in inventories, scaled by lagged total assets, $(\Delta PPEGT + \Delta INVT)/AT_{t-1}$, following Lyandres et al. (2008). Forward-filled monthly with the four-month reporting lag. Sign-flipped.
- **Net Stock Issues (NSI).** Annual log change in split-adjusted shares outstanding, $\log(adj_shares_t/adj_shares_{t-1})$, following Fama and French (2008). Stocks with negative NSI are placed in decile 1, stocks with zero NSI in decile 2, and stocks with positive NSI are sorted into deciles 3–10 using NYSE breakpoints from the positive sub-sample. The signal is then sign-flipped (high-issuance firms earn lower expected returns).
- **Gross Profitability (GP).** Annual gross profits scaled by total assets, $(REVT - COGS)/AT$, following Novy-Marx (2013). Forward-filled monthly with the four-month reporting lag.
- **Accruals (Accrual).** Change in non-cash working capital less depreciation, scaled by average total assets, $(\Delta NCWC - DP)/\bar{AT}$, following Sloan (1996). Forward-filled monthly with the four-month reporting lag. Sign-flipped.
- **Net Operating Assets (NOA).** Operating assets less operating liabilities, scaled by lagged total assets, following Hirshleifer et al. (2004). Forward-filled monthly with the four-month reporting lag. Sign- flipped.
- **Ohlson’s O-Score (OScore).** Computed according to the nine-variable specification of Ohlson (1980) using annual Compustat data. Forward-filled monthly with the four-month reporting lag. Sign-flipped.
- **Return on Equity (ROE).** Income before extraordinary items from quarterly Compustat (IBQ) divided by one-quarter-lagged quarterly book equity. Each observation is dated four months after `datadate` and forward-filled monthly onto the CRSP panel until the next quarterly observation becomes available. Financial firms (SIC 6000–6999) are excluded.
- **Quarterly Return on Assets (ROA).** IBQ_t/ATQ_{t-1} , following Chen et al. (2010). Updated monthly with RDQ-based timing.
- **Failure Probability (FailProb).** Computed according to Campbell et al. (2008). Inputs are constructed from CRSP and the merged CRSP-Compustat file, cross-sectionally winsorized at the 5th and 95th percentiles each month, and combined using the published linear coefficients. Following Stambaugh and Yuan (2017), stocks with a price below \$5 are excluded. Updated monthly. Sign-flipped.
- **Composite Equity Issuance, 12-month (CoEqIs12).** As in Daniel and Titman (2006), computed each month from CRSP as $\log(ME_t/ME_{t-12})$ less the 12-month cumulative log return, following the one-year horizon used in Stambaugh and Yuan (2017). Lagged four months to align with the timing of the annual Compustat signals. Sign-flipped.

Signals taken from Chen and Zimmermann (2022)

The following four signals are taken directly from the open-source characteristic library of Chen and Zimmermann (2022). We use their provided values without modification and merge each characteristic onto our CRSP panel by `permno` and `yyyymm`. Chen and Zimmermann sign each predictor so that high values correspond to high expected returns, so no sign flip is needed.

- **Beta (Beta).** Market beta from a 60-month rolling time-series regression of stock returns on the market return. osap signal name: Beta.
- **Book-to-Market (B/M).** Book equity for the fiscal year ending in calendar year $t - 1$ scaled by market equity at the end of December of year $t - 1$. The construction process follows Fama and French (1993). osap signal name: BMdec.
- **Announcement Return (AnnRet).** Three-day cumulative return centered on the most recent quarterly earnings-announcement date, following Chan et al. (1996). osap signal name: AnnouncementReturn. We use this as the empirical counterpart to the post-earnings-announcement-drift (PEAD) leg of the Daniel et al. (2020) three-factor model.
- **Composite Equity Issuance, 5-year (CoEqIs).** Five-year composite equity issuance. osap signal name: CompEquIss.

Table A1: **Expected Skewness and Factor Returns Summary Statistics**

This table reports summary statistics for the variables used in the empirical analysis. Panel A reports time-series averages of monthly cross-sectional summary statistics for the expected skewness and expected idiosyncratic skewness used to form factor portfolios. Panel B reports time-series summary statistics of monthly long-short factor returns, in percent per month, for the raw (Standard) and skewness-managed versions of each factor. The sample is monthly from July 1963 to December 2024.

Panel A: Expected Skewness							
	mean	std	min	25%	50%	75%	max
Expected skewness	0.24	0.21	-0.82	0.10	0.23	0.37	1.60
Expected idio. skewness	0.19	0.17	-0.65	0.09	0.19	0.29	1.28
Panel B: Factor Returns (% per month)							
	mean	std	min	25%	50%	75%	max
Beta (Standard)	0.19	7.89	-28.95	-4.49	-0.10	4.55	43.02
Beta (Skewness Managed)	0.74	10.84	-36.67	-5.24	0.19	5.87	96.38
Size (Standard)	0.12	4.79	-14.19	-2.92	-0.14	2.56	27.99
Size (Skewness Managed)	0.50	7.73	-24.69	-3.54	0.10	3.77	74.82
B/M (Standard)	0.28	4.49	-14.63	-2.56	0.04	2.91	21.48
B/M (Skewness Managed)	1.09	7.05	-27.57	-2.88	0.90	4.67	39.78
OP (Standard)	0.35	4.46	-21.73	-1.89	0.37	2.62	24.81
OP (Skewness Managed)	0.93	5.43	-21.42	-2.16	0.92	3.93	29.83
Inv (Standard)	0.41	3.59	-12.02	-1.95	0.47	2.33	20.67
Inv (Skewness Managed)	1.17	7.08	-30.58	-2.36	0.66	3.77	72.63
Mom (Standard)	1.16	7.17	-45.33	-1.58	1.70	5.02	23.78
Mom (Skewness Managed)	1.53	7.82	-37.14	-2.28	2.05	5.67	43.42
ROE (Standard)	0.50	5.05	-28.70	-2.21	0.55	3.08	21.47
ROE (Skewness Managed)	0.92	5.79	-21.46	-2.60	0.99	4.48	27.89
CoEqIs (Standard)	0.48	4.07	-17.79	-1.84	0.66	2.95	14.24
CoEqIs (Skewness Managed)	0.80	7.46	-28.18	-2.46	0.91	3.72	106.77
AnnRet (Standard)	0.78	3.31	-12.64	-1.09	0.70	2.62	16.00
AnnRet (Skewness Managed)	1.37	6.24	-43.41	-1.88	1.00	3.95	56.46
NSI (Standard)	0.53	2.86	-10.97	-1.13	0.54	2.16	13.19
NSI (Skewness Managed)	0.91	4.16	-15.94	-1.24	0.99	3.19	30.72
CoEqIs12 (Standard)	0.43	3.61	-13.87	-1.44	0.36	2.51	14.48
CoEqIs12 (Skewness Managed)	1.15	4.67	-18.74	-1.53	0.98	3.72	31.95
Accrual (Standard)	0.31	3.44	-12.93	-1.62	0.23	2.26	13.48
Accrual (Skewness Managed)	0.69	7.35	-28.13	-2.86	0.44	3.86	52.87
NOA (Standard)	0.53	2.96	-14.82	-1.40	0.51	2.46	10.73
NOA (Skewness Managed)	0.59	6.20	-34.40	-2.44	0.44	3.36	40.79
InvA (Standard)	0.46	3.51	-15.22	-1.66	0.47	2.56	20.93
InvA (Skewness Managed)	0.92	7.01	-26.61	-2.71	0.84	3.93	50.47
FailProb (Standard)	0.50	7.34	-32.81	-2.91	0.85	4.66	32.47
FailProb (Skewness Managed)	0.67	7.71	-34.01	-3.27	1.19	5.09	31.30
OScore (Standard)	0.12	4.19	-17.65	-2.28	-0.03	2.71	15.65
OScore (Skewness Managed)	0.48	5.41	-18.66	-2.48	0.09	3.38	28.91
GP (Standard)	0.28	3.84	-13.13	-2.02	0.27	2.59	18.72
GP (Skewness Managed)	0.89	5.99	-31.84	-2.25	0.96	3.94	35.92
ROA (Standard)	0.64	4.65	-19.65	-1.89	0.56	3.31	22.63
ROA (Skewness Managed)	0.91	5.56	-21.98	-2.29	0.85	4.24	21.63

Table A2: **Average Realized Skewness by Portfolio**

Portfolios are first sorted on the anomaly characteristic and then merged with stock-level expected skewness. The second sort is done sequentially within each characteristic portfolio, yielding 30 portfolios in total. Cell entries are average realized monthly skewness, value-weighted by prior-month market cap. The H-L column reports the within-port skew gap (high-low predicted skewness); the bottom row reports the across-port skew gap; the bottom-right corner reports the diagonal (long/high – short/low for Panel A, short/high – long/low for Panel B). Newey-West HAC t-statistics with 12 monthly lags appear in parentheses below each L-S / S-L mean.

Panel A: Skewness in Long Leg

B/M					Size					Inv				
port	high	med	low	H-L	port	high	med	low	H-L	port	high	med	low	H-L
Long	0.29	0.22	0.16	0.13	Long	0.39	0.30	0.22	0.16	Long	0.33	0.21	0.16	0.17
9	0.23	0.17	0.14	0.10	9	0.32	0.26	0.21	0.11	9	0.20	0.16	0.15	0.05
8	0.20	0.16	0.11	0.09	8	0.28	0.23	0.19	0.09	8	0.17	0.15	0.10	0.07
7	0.17	0.15	0.11	0.07	7	0.25	0.21	0.17	0.08	7	0.19	0.14	0.11	0.07
6	0.17	0.13	0.11	0.06	6	0.23	0.19	0.15	0.07	6	0.15	0.12	0.09	0.06
5	0.17	0.14	0.10	0.08	5	0.21	0.17	0.15	0.06	5	0.14	0.13	0.09	0.05
4	0.18	0.13	0.09	0.09	4	0.20	0.15	0.12	0.07	4	0.15	0.11	0.10	0.06
3	0.15	0.12	0.10	0.05	3	0.17	0.14	0.10	0.07	3	0.18	0.12	0.09	0.09
2	0.17	0.13	0.10	0.07	2	0.14	0.12	0.09	0.05	2	0.18	0.13	0.11	0.07
Short	0.21	0.13	0.11	0.10	Short	0.12	0.11	0.09	0.03	Short	0.22	0.15	0.11	0.10
L-S	0.09	0.10	0.06	0.19	L-S	0.27	0.20	0.13	0.30	L-S	0.11	0.06	0.05	0.22
	(5.01)	(6.42)	(3.70)	(10.68)		(19.68)	(14.71)	(8.05)	(22.02)		(5.73)	(4.40)	(3.41)	(12.08)

Panel B: Skewness in Short Leg

MOM					ROE					OP				
port	high	med	low	H-L	port	high	med	low	H-L	port	high	med	low	H-L
Short	0.34	0.27	0.21	0.12	Short	0.34	0.25	0.19	0.16	Short	0.37	0.26	0.19	0.18
2	0.24	0.22	0.16	0.08	2	0.26	0.20	0.14	0.12	2	0.23	0.18	0.13	0.10
3	0.22	0.17	0.15	0.07	3	0.21	0.17	0.12	0.09	3	0.19	0.14	0.11	0.08
4	0.20	0.17	0.13	0.08	4	0.17	0.16	0.10	0.06	4	0.18	0.15	0.12	0.06
5	0.17	0.13	0.10	0.07	5	0.15	0.12	0.09	0.06	5	0.18	0.15	0.11	0.07
6	0.16	0.13	0.10	0.06	6	0.14	0.14	0.08	0.06	6	0.15	0.12	0.10	0.05
7	0.16	0.12	0.08	0.08	7	0.15	0.10	0.08	0.07	7	0.15	0.12	0.08	0.07
8	0.15	0.10	0.09	0.06	8	0.12	0.10	0.09	0.03	8	0.15	0.12	0.09	0.06
9	0.15	0.11	0.09	0.06	9	0.12	0.09	0.08	0.04	9	0.15	0.12	0.07	0.08
Long	0.20	0.13	0.11	0.09	Long	0.14	0.11	0.08	0.05	Long	0.18	0.12	0.09	0.09
S-L	0.14	0.14	0.11	0.23	S-L	0.21	0.14	0.11	0.26	S-L	0.19	0.14	0.10	0.28
	(6.79)	(7.74)	(5.41)	(10.87)		(10.57)	(10.20)	(7.31)	(14.81)		(9.76)	(9.91)	(7.01)	(16.15)

Table A3: Scaled Skewness-Managed Strategy Performance

This table details the outcome when managing skewness via the scaling strategy. Exposure to the anomaly is scaled by average expected skewness in the long and short legs. Strategies in which skewness dominates the long leg increase exposure as expected skewness rises, while those in which skewness dominates the short leg decrease exposure. Panel A reports the performance of the original anomaly portfolio in the sample where both the characteristic and the expected skewness measure are non-missing. Panel B reports the better-performing scaled version per anomaly: “Long” applies the scalar to the long leg, “Short” to the short leg; the choice is the version with the higher Sharpe ratio. Panel C reports the difference (Panel B – Panel A): the t-stat under Mean is a Newey-West HAC t (lag 12), the z-score under Sharpe is the Jobson-Korkie (1981) test, and the t-stat under Skew is from a paired moving-block bootstrap (block = 12, B = 2000).

	Beta	Size	B/M	OP	Inv	Mom	ROE	CoEqIs	AnnRet	NSI	CoEqIs12	Accrual	NOA	InvA	FailProb	OScore	GP	ROA
Panel A: Standard Anomaly																		
mean	2.29	1.44	3.35	4.21	4.93	13.90	6.70	5.74	9.25	6.40	5.19	3.69	6.37	5.57	3.96	1.45	3.36	6.86
sharpe	0.08	0.09	0.22	0.27	0.40	0.56	0.37	0.41	0.81	0.65	0.42	0.31	0.62	0.46	0.16	0.10	0.25	0.41
skew	0.50	0.84	0.36	0.23	0.48	-1.42	-0.18	-0.26	0.17	-0.02	-0.03	-0.05	-0.18	0.15	-0.28	-0.21	0.20	-0.11
Panel B: Best Skewness Scaling Strategy																		
Strategy	Long	Long	Long	Short	Long	Short	Short	Short	Short	Short	Short	Long	Short	Long	Short	Short	Short	Short
mean	10.25	10.50	5.65	6.66	5.42	16.54	11.49	6.50	10.35	8.62	6.81	3.50	7.08	5.59	5.99	5.24	3.96	9.06
sharpe	0.33	0.55	0.34	0.36	0.39	0.55	0.56	0.36	0.74	0.70	0.44	0.28	0.56	0.47	0.19	0.33	0.23	0.53
skew	2.19	2.04	1.09	0.21	0.95	-1.03	0.22	-0.32	0.76	0.27	-0.29	0.43	-0.21	0.39	-0.44	-0.12	0.64	-0.07
Panel C: Difference																		
Mean	7.96	9.06	2.31	2.44	0.49	2.64	4.78	0.76	1.10	2.23	1.62	-0.19	0.71	0.03	1.57	3.80	0.60	2.44
t-stat	(3.98)	(6.59)	(1.96)	(2.10)	(0.53)	(1.34)	(3.79)	(0.70)	(1.20)	(3.09)	(1.73)	(-0.23)	(0.99)	(0.03)	(0.81)	(3.25)	(0.57)	(1.80)
Sharpe	0.25	0.47	0.12	0.09	-0.01	-0.01	0.19	-0.05	-0.07	0.05	0.02	-0.03	-0.07	0.02	0.03	0.23	-0.02	0.12
zscore	(3.60)	(6.55)	(1.69)	(1.49)	(-0.08)	(-0.15)	(2.86)	(-0.81)	(-0.91)	(0.88)	(0.35)	(-0.46)	(-1.08)	(0.20)	(0.47)	(3.16)	(-0.30)	(1.61)
Skew	1.70	1.20	0.73	-0.22	0.80	0.39	0.14	-0.06	0.59	0.29	-0.26	-0.07	0.09	0.23	-0.15	0.09	0.79	-0.65
t-stat	(3.90)	(5.03)	(1.99)	(-0.64)	(1.98)	(1.30)	(0.34)	(-0.19)	(1.27)	(1.46)	(-1.10)	(-0.18)	(0.55)	(0.81)	(-0.73)	(0.34)	(2.21)	(-0.85)

Table A4: **Transaction-Cost Summary Statistics**

This table reports summary statistics for the one-way effective spread estimates from the Hasbrouck (2009) Gibbs sampler in the CRSP common-stock universe from July 1963 to December 2024 (738 months). The cost variable c is expressed as a decimal fraction of transaction value (cost per dollar transacted). Panel A reports the time-series average of monthly cross-sectional distribution statistics: for each month we compute the mean, standard deviation, and percentiles of c across all stocks with a valid Gibbs estimate, then average each statistic across the 738 months. The average number of stocks per month with a valid cost estimate is 3,917. Panel B reports the time-series average of monthly transaction-cost statistics, computing equal-weighted (EW, per stock) and value-weighted (VW, by market capitalization) versions of three statistics: (i) the share of stocks for which the Gibbs cost estimate is missing, (ii) the average cost across stocks with a valid estimate, and (iii) the share with $c > 0.01$ (one percent of transaction value). Sub-periods are defined by data-availability regime: NASDAQ stocks enter the CRSP daily file in December 1972, the NASDAQ National Market System (NMS) launched in April 1982, was fully extended in June 1992, and U.S. exchanges completed the transition to decimal pricing in April 2001.

Panel A: Cross-Sectional Distribution of c							
	Mean	Std Dev	Min	P25	Median	P75	Max
c	0.0098	0.0106	0.0006	0.0033	0.0063	0.0123	0.1338

Panel B: Sub-Period Averages							
Sub-period	Months	Missing c (%)		Avg c		$c > 0.01$ (%)	
		EW	VW	EW	VW	EW	VW
Full sample (07/1963–12/2024)	738	13.80	2.02	0.0098	0.0028	22.90	1.82
1963–1971 (pre-NASDAQ)	102	0.89	0.18	0.0072	0.0026	19.16	1.13
1972–1981 (NASDAQ in CRSP, no NMS)	120	45.80	8.90	0.0101	0.0028	16.90	1.85
1982–1991 (NMS rollout)	120	33.51	2.76	0.0138	0.0031	28.72	3.18
1992–2024 (post-full-NMS)	396	1.45	0.18	0.0092	0.0027	23.92	1.57
1992–2002 (pre-decimalization)	132	2.74	0.26	0.0168	0.0040	48.19	4.19
2003–2024 (decimalization era)	264	0.80	0.14	0.0054	0.0020	11.78	0.26

Table A5: **Skewness Management Strategy Performance, Net of Transaction Costs**

This table compares the performance of the original anomaly portfolios with their skewness-managed counterparts, after accounting for transaction costs. Panel A reports the after-cost performance of the original factor portfolios in the subsample where the estimated round-trip transaction costs are below 0.01 and both the characteristic signal and the expected-skewness measure are non-missing. Panel B reports the after-cost performance of the skewness-managed portfolios, constructed by selecting stocks with high expected skewness for the long leg and low expected skewness for the short leg of each anomaly. Panel C reports the difference between the two. The Newey-West t -statistic on the mean difference (lag 12), the Jobson-Korkie (1981) z -statistic on the Sharpe-ratio difference, and the paired moving-block bootstrap t -statistic on the skewness difference are reported in parentheses below each row.

	Beta	Size	B/M	OP	Inv	Mom	ROE	CoEqIs	AnnRet	NSI	CoEqIs12	Accrual	NOA	InvA	FailProb	OScore	GP	ROA
Panel A: Standard Anomaly (Net of Costs)																		
Mean	3.73	12.16	3.27	1.24	4.79	8.47	1.92	4.52	7.39	5.07	3.84	3.15	5.52	4.94	0.48	-3.60	2.36	3.72
Sharpe	0.14	0.75	0.21	0.08	0.38	0.34	0.11	0.31	0.63	0.52	0.31	0.26	0.53	0.40	0.02	-0.26	0.17	0.24
Skew	0.42	0.75	0.27	0.28	0.39	-1.45	0.00	-0.30	0.14	-0.07	-0.03	-0.05	-0.18	0.15	-0.33	0.14	0.12	-0.03
Panel B: Skewness Managed (Net of Costs)																		
Mean	8.30	18.38	9.70	3.47	11.12	7.70	1.09	3.81	11.88	5.43	7.04	4.45	2.70	6.12	0.25	-5.67	3.19	2.29
Sharpe	0.22	0.66	0.37	0.18	0.41	0.28	0.05	0.13	0.53	0.38	0.44	0.17	0.12	0.24	0.01	-0.30	0.15	0.11
Skew	1.53	3.04	0.81	0.27	4.32	-0.51	0.04	7.36	0.91	0.35	0.08	1.07	0.17	1.53	-0.48	0.22	0.19	0.04
Panel C: Difference																		
Mean	4.56	6.22	6.43	2.23	6.36	-0.77	-0.82	-0.64	4.49	0.36	3.20	1.30	-2.82	1.19	-0.24	-2.07	0.82	-1.42
t-stat	(1.62)	(2.64)	(2.53)	(1.16)	(1.99)	(-0.37)	(-0.36)	(-0.18)	(1.50)	(0.21)	(1.62)	(0.39)	(-0.96)	(0.49)	(-0.13)	(-0.98)	(0.32)	(-0.68)
Sharpe	0.08	-0.10	0.16	0.10	0.03	-0.06	-0.06	-0.18	-0.10	-0.14	0.13	-0.09	-0.41	-0.15	-0.01	-0.04	-0.02	-0.12
zscore	(1.22)	(-1.22)	(1.36)	(0.86)	(0.18)	(-0.71)	(-0.48)	(-1.33)	(-0.64)	(-1.06)	(1.00)	(-0.61)	(-2.56)	(-1.10)	(-0.14)	(-0.29)	(-0.15)	(-0.98)
Skew	1.10	2.30	0.54	-0.01	3.93	0.94	0.03	7.67	0.77	0.41	0.11	1.13	0.35	1.38	-0.16	0.07	0.07	0.07
t-stat	(2.12)	(3.09)	(2.06)	(-0.02)	(2.17)	(3.26)	(0.14)	(2.01)	(1.49)	(0.92)	(0.51)	(3.54)	(1.01)	(2.62)	(-1.33)	(0.29)	(0.14)	(0.32)

Table A6: **Spanning Tests with $E[Skew]$ Anomaly**

This table reports spanning regressions of the skewness-managed long-short returns on benchmarks built from the expected-skewness anomaly. The $E[Skew]$ portfolio is a value-weighted long-short return on stocks sorted into terciles by predicted skewness. Panel A regresses each skewness-managed long-short return on $E[Skew]$ alone. Panel B adds the corresponding original anomaly portfolio. Panel C further adds the Fama-French five-factor model. Alpha is annualized in percent. Newey-West (lag 12) t -statistics in parentheses. The t -statistics for β_{orig} and $\beta_{E[Skew]}$ test $H_0 : \beta = 1$; alpha t -statistics test against zero. R^2 is adjusted R^2 .

	Beta	Size	B/M	OP	Inv	Mom	ROE	CoEqIs	AnnRet	NSI	CoEqIs12	Accrual	NOA	InvA	FailProb	OScore	GP	ROA
Panel A: $E[Skew]$ only																		
α	3.18 (0.87)	1.86 (0.64)	10.42 (3.16)	10.60 (3.78)	10.88 (3.57)	17.35 (5.76)	10.45 (3.44)	6.41 (2.51)	14.41 (5.90)	9.34 (5.17)	12.79 (5.56)	4.77 (1.80)	4.31 (1.88)	8.49 (2.68)	8.64 (2.47)	4.62 (1.75)	8.12 (3.62)	10.14 (3.28)
$\beta_{E[Skew]}$	2.16 (7.20)	1.56 (5.97)	1.02 (0.15)	0.22 (-5.11)	1.17 (0.93)	0.38 (-2.05)	0.29 (-6.75)	1.24 (1.40)	0.97 (-0.18)	0.59 (-3.40)	0.37 (-6.16)	1.32 (2.72)	1.03 (0.17)	1.01 (0.11)	-0.30 (-4.45)	0.54 (-5.42)	0.96 (-0.29)	0.38 (-6.28)
R^2	0.43	0.44	0.22	0.02	0.29	0.02	0.03	0.30	0.28	0.22	0.07	0.34	0.29	0.22	0.02	0.11	0.28	0.05
Panel B: original + $E[Skew]$																		
α	4.43 (2.07)	1.88 (1.51)	7.62 (3.88)	5.07 (3.06)	6.80 (2.91)	3.40 (2.01)	3.93 (1.89)	2.49 (1.03)	6.82 (2.69)	2.29 (2.25)	7.67 (4.63)	2.17 (0.92)	-0.03 (-0.02)	3.50 (1.55)	0.64 (0.36)	2.75 (1.53)	5.45 (2.91)	1.97 (1.05)
β_{orig}	1.02 (0.31)	1.15 (3.31)	0.90 (-2.03)	0.98 (-0.41)	0.77 (-3.17)	0.93 (-1.79)	0.90 (-1.98)	0.74 (-4.11)	0.80 (-2.00)	0.98 (-0.47)	0.83 (-3.31)	0.70 (-4.12)	0.65 (-4.87)	0.88 (-1.59)	1.01 (0.41)	0.86 (-3.17)	0.73 (-6.21)	0.93 (-1.94)
$\beta_{E[Skew]}$	0.81 (-2.67)	0.92 (-1.04)	0.93 (-0.95)	0.75 (-3.94)	1.27 (1.45)	0.75 (-3.96)	0.89 (-1.46)	1.12 (0.73)	1.03 (0.21)	0.89 (-1.16)	0.67 (-5.04)	1.32 (2.77)	1.11 (0.71)	1.04 (0.49)	0.59 (-2.59)	0.85 (-1.96)	1.04 (0.32)	0.85 (-1.60)
R^2	0.81	0.88	0.55	0.57	0.44	0.73	0.53	0.46	0.46	0.62	0.44	0.45	0.39	0.42	0.79	0.52	0.58	0.53
Panel C: FF5 + original + $E[Skew]$																		
α	3.66 (1.78)	3.73 (2.48)	5.09 (2.66)	3.15 (2.27)	9.18 (3.72)	2.77 (1.63)	3.08 (1.64)	-0.44 (-0.18)	7.74 (3.10)	1.31 (1.13)	6.71 (3.96)	2.08 (0.73)	3.31 (1.64)	1.48 (0.69)	1.85 (1.04)	1.21 (0.62)	6.37 (3.52)	1.69 (1.00)
β_{orig}	0.97 (-0.27)	1.39 (3.38)	0.42 (-6.12)	0.93 (-1.09)	0.68 (-3.62)	0.94 (-1.58)	0.97 (-0.48)	0.77 (-3.14)	0.77 (-2.80)	0.98 (-0.35)	0.72 (-4.13)	0.72 (-3.31)	0.72 (-4.43)	0.72 (-3.38)	1.09 (1.64)	1.06 (0.82)	0.70 (-4.36)	0.95 (-0.93)
$\beta_{E[Skew]}$	0.79 (-2.10)	0.79 (-3.54)	0.76 (-2.48)	0.75 (-3.99)	0.88 (-1.07)	0.66 (-5.66)	0.87 (-1.48)	1.06 (0.30)	0.87 (-1.52)	0.87 (-1.85)	0.68 (-5.24)	1.10 (1.13)	0.71 (-3.83)	0.88 (-1.49)	0.47 (-7.84)	0.79 (-2.86)	0.87 (-1.23)	0.87 (-1.38)
R^2	0.81	0.90	0.61	0.58	0.54	0.74	0.53	0.47	0.49	0.63	0.45	0.49	0.51	0.47	0.81	0.53	0.52	0.58

Table A7: **Predictive Regressions for Idiosyncratic Skewness**

This table reports Fama–MacBeth (1973) estimates from monthly cross-sectional regressions of next-month idiosyncratic realized skewness on the predictors listed below. The dependent variable is monthly idiosyncratic realized skewness (idioskew_t), and the realized-skewness (rs) and realized-volatility (rv) predictors are computed from daily idiosyncratic returns. For each model we run a separate cross-sectional regression in every month and then average the resulting coefficients, adjusted R^2 , and number of observations over time. The row labeled *Avg.* reports the time-series average of the monthly cross-sectional coefficients; the row labeled *%Sig* reports the fraction of months in which the monthly coefficient is statistically significant at the 5% level and shares the sign of the time-series average. The sample is the universe of common stocks trading on AMEX, NYSE, and Nasdaq and is constructed so that the resulting expected idiosyncratic skewness estimates span July 1963 through December 2024, the period used to form the skewness-managed portfolios. Momentum (mom) is the prior-year return omitting the most recent month and prior is the most recent month’s return. Dummy variables are included for NASDAQ stocks, stocks in the small (sm) and medium (med) size terciles of market capitalization, and Fama–French 48 industries (ind). R^2 is the time-series average of the monthly cross-sectional adjusted R^2 , and N is the average number of stocks per monthly cross-section.

Model		rv_t-1	rs_t-1	mom	turnover	nasdaq	sm	med	prior	ind	R^2	N
1	Avg.	2.18	0.01							No	0.006	4489
	%Sig	(0.75)	(0.28)									
2	Avg.	1.65	0.01	-0.01	0.11	-0.01	0.09	0.07		No	0.012	3835
	%Sig	(0.59)	(0.19)	(0.19)	(0.22)	(0.24)	(0.58)	(0.54)				
3	Avg.									yes	0.007	4474
	%Sig											
4	Avg.	1.46	0.01	-0.01	0.08	-0.02	0.09	0.07		yes	0.018	3810
	%Sig	(0.53)	(0.18)	(0.18)	(0.18)	(0.26)	(0.56)	(0.52)				
5	Avg.								-0.16	no	0.001	4517
	%Sig								(0.46)			
6	Avg.	1.70	0.02	-0.01	0.12	-0.02	0.08	0.06	-0.24	yes	0.019	3810
	%Sig	(0.58)	(0.32)	(0.18)	(0.21)	(0.26)	(0.54)	(0.50)	(0.61)			
7	Avg.	1.76	0.02	-0.01		-0.02	0.07	0.06	-0.26	yes	0.020	4078
	%Sig	(0.64)	(0.35)	(0.17)		(0.29)	(0.54)	(0.48)	(0.66)			

Table A8: **Skewness Management Strategy Performance (Idiosyncratic Skewness)**

This table compares the original to the idiosyncratic-skewness-managed versions of each anomaly. Expected skewness is the predicted value from the cross-sectional regression model for next-month idiosyncratic realized skewness. Panel A lists performance in the full sample in which both the characteristic and expected idiosyncratic skewness are non-missing. Panel B lists the performance of the skewness-managed version of each anomaly in which the long leg comes from the high expected idiosyncratic skewness sample and the short leg comes from the low expected idiosyncratic skewness sample. Panel C reports the difference. The Newey-West t -statistic on the mean difference (lag 12), the Jobson and Korkie (1981) z -statistic on the Sharpe-ratio difference, and the paired moving-block bootstrap t -statistic on the skewness difference are reported in parentheses.

	Beta	Size	B/M	OP	Inv	Mom	ROE	CoEqIs	AnnRet	NSI	CoEqIs12	Accrual	NOA	InvA	FailProb	OScore	GP	ROA
Panel A: Standard Anomaly																		
Mean	2.29	1.44	3.35	4.21	4.93	13.90	5.96	5.74	9.36	6.40	5.19	3.69	6.37	5.57	6.05	1.43	3.36	7.68
Sharpe	0.08	0.09	0.21	0.27	0.40	0.56	0.34	0.41	0.82	0.65	0.42	0.31	0.62	0.46	0.24	0.10	0.25	0.48
Skew	0.50	0.84	0.36	0.23	0.48	-1.42	-0.28	-0.26	0.16	-0.02	-0.03	-0.05	-0.18	0.15	-0.33	-0.21	0.20	-0.06
Panel B: Idiosyncratic Skewness Managed																		
Mean	7.27	7.52	13.28	11.62	10.81	15.94	11.14	10.86	14.09	12.28	13.23	8.05	9.61	10.62	9.01	4.65	10.61	12.61
Sharpe	0.20	0.29	0.53	0.62	0.52	0.58	0.54	0.46	0.64	0.86	0.78	0.31	0.46	0.48	0.32	0.27	0.56	0.64
Skew	1.76	1.85	1.08	0.44	0.45	-0.66	0.32	1.68	1.05	0.41	0.73	0.94	0.88	0.10	-0.21	-0.05	0.17	0.16
Panel C: Difference																		
Mean	4.98	6.07	9.93	7.41	5.88	2.04	5.19	5.12	4.73	5.88	8.04	4.36	3.24	5.05	2.96	3.22	7.24	4.93
t-stat	(1.78)	(3.27)	(3.53)	(3.49)	(2.29)	(0.99)	(2.24)	(1.93)	(1.66)	(3.15)	(4.12)	(1.53)	(1.31)	(2.26)	(1.78)	(1.34)	(3.00)	(2.01)
Sharpe	0.11	0.21	0.31	0.34	0.12	0.02	0.20	0.05	-0.18	0.22	0.37	0.00	-0.16	0.02	0.08	0.17	0.31	0.17
zscore	(1.67)	(3.11)	(2.76)	(3.30)	(0.90)	(0.23)	(1.84)	(0.43)	(-1.16)	(1.74)	(3.23)	(0.03)	(-1.03)	(0.18)	(1.17)	(1.24)	(2.43)	(1.46)
Skew	1.27	1.01	0.72	0.21	-0.03	0.76	0.60	1.95	0.88	0.43	0.76	0.99	1.07	-0.05	0.12	0.16	-0.02	0.22
t-stat	(2.55)	(1.97)	(1.06)	(0.77)	(-0.08)	(3.43)	(2.14)	(1.65)	(1.98)	(1.40)	(1.51)	(2.34)	(3.41)	(-0.07)	(0.70)	(0.86)	(-0.05)	(1.11)

Table A9: **Predictive Regressions without Continuous Return Predictors**

This table reports the time-series average estimates from monthly cross-sectional regressions of realized skewness on subsets of the predictors in the benchmark Model 7 of Table 3. Model 8 drops realized volatility (rv_t-1) and turnover from the benchmark; Model 9 further drops momentum (mom) and prior. For each model we run a separate cross-sectional regression in every month and then average the resulting coefficients, adjusted R^2 , and number of observations over time. The row labeled *Avg.* reports the time-series average of the monthly cross-sectional coefficients; the row labeled *%Sig* reports the fraction of months in which the monthly coefficient is statistically significant at the 5% level and shares the sign of the time-series average. The sample is the universe of common stocks trading on AMEX, NYSE, and Nasdaq and is constructed so that the resulting expected skewness estimates span July 1963 through December 2024, the period used to form the skewness-managed portfolios. Momentum (mom) is the prior-year return omitting the most recent month and prior is the most recent month's return. Dummy variables are included for NASDAQ stocks, stocks in the small (sm) and medium (med) size terciles, and Fama–French 48 industries (ind). R^2 is the time-series average of the monthly cross-sectional adjusted R^2 , and N is the average number of stocks per monthly cross-section.

Model	rv_t-1	rs_t-1	mom	turnover	nasdaq	sm	med	prior	ind	R^2	N
8 Avg.		0.03	-0.03		-0.02	0.14	0.09	-0.34	yes	0.027	4139
%Sig		(0.51)	(0.30)		(0.27)	(0.64)	(0.60)	(0.65)			
9 Avg.		0.02			-0.03	0.15	0.10		yes	0.023	4518
%Sig		(0.35)			(0.30)	(0.68)	(0.63)				

Table A10: **Skewness Managed Performance without Continuous Return Predictors**

This table compares the original anomaly portfolios to skewness-managed versions where expected skewness is forecast without certain continuous return predictors. Panel A is the standard anomaly in the sample where both the characteristic and the alternative expected-skewness measure are non-missing. Panel B drops realized volatility from the predictor set (model 8). Panel C drops realized volatility, momentum, and prior-month return (model 9).

	Beta	Size	B/M	OP	Inv	Mom	ROE	CoEqIs	AnnRet	NSI	CoEqIs12	Accrual	NOA	InvA	FailProb	OScore	GP	ROA
Panel A: Standard Anomaly																		
mean	2.29	1.44	3.35	4.21	4.93	13.90	5.96	5.74	9.36	6.40	5.19	3.69	6.37	5.57	6.05	1.43	3.36	7.68
sharpe	0.08	0.09	0.21	0.27	0.40	0.56	0.34	0.41	0.82	0.65	0.42	0.31	0.62	0.46	0.24	0.10	0.25	0.48
skew	0.50	0.84	0.36	0.23	0.48	-1.42	-0.28	-0.26	0.16	-0.02	-0.03	-0.05	-0.18	0.15	-0.33	-0.21	0.20	-0.06
Panel B: Skewness Managed Less Realized Volatility (model 8)																		
mean	9.44	8.10	13.57	12.38	13.47	18.26	12.10	10.80	15.91	11.61	12.01	8.69	9.42	11.11	9.59	5.97	10.64	11.80
sharpe	0.26	0.32	0.57	0.66	0.57	0.67	0.59	0.39	0.78	0.78	0.75	0.36	0.45	0.49	0.36	0.33	0.51	0.61
skew	1.47	2.13	0.64	-0.12	1.17	-0.75	-0.04	4.99	0.64	0.06	0.10	1.53	0.78	1.20	-0.45	0.11	0.54	-0.19
Panel C: Skewness Managed Less Realized Volatility, Momentum, Prior Month Return (model 9)																		
mean	7.53	5.58	10.37	12.58	10.69	15.30	9.23	5.78	14.65	11.64	12.13	7.14	7.54	11.01	10.35	5.99	11.19	12.63
sharpe	0.22	0.24	0.47	0.67	0.50	0.53	0.44	0.28	0.69	0.76	0.77	0.32	0.38	0.50	0.38	0.33	0.56	0.64
skew	1.11	1.57	0.37	0.30	1.19	-0.50	-0.21	0.25	1.85	0.14	0.11	0.59	1.22	1.29	-0.20	0.05	0.45	-0.03

Table A11: **Macroeconomic State Variables: Summary Statistics and Pairwise Correlations**

Panel A reports time-series summary statistics for the six macroeconomic state variables used in the time-series analysis. Panel B reports their pairwise Pearson correlations. The sample spans July 1963 through December 2024 for all variables except VIX, which is available from January 1990 onward. Recession is the NBER binary recession indicator. Credit Spread is the Baa–Aaa corporate bond yield spread. CS Disp (IQR) is the interquartile range of the cross-sectional distribution of monthly stock returns. VIX is the CBOE implied volatility index. $F1^3$ is the nonlinear real activity factor of Ludvigson and Ng (2007). Market Return is the CRSP value-weighted excess return.

Panel A: Summary Statistics							
	mean	std	min	25%	50%	75%	max
(1) Recession	0.12	0.32	0	0	0	0	1
(2) Credit Spread	1.02	0.43	0.32	0.73	0.91	1.19	3.38
(3) CS Disp (IQR)	13.06	3.51	6.70	10.75	12.43	14.36	36.52
(4) VIX	19.47	7.49	10.13	14.02	17.67	23.15	62.67
(5) $F1^3$	0.07	0.63	-3.30	-0.01	0.00	0.01	8.96
(6) Market Return	0.59	4.48	-23.19	-1.96	0.98	3.44	16.11
Panel B: Pairwise Correlations							
	(1)	(2)	(3)	(4)	(5)	(6)	
(1) Recession	1.00						
(2) Credit Spread	0.44	1.00					
(3) CS Disp (IQR)	0.32	0.31	1.00				
(4) VIX	0.50	0.63	0.64	1.00			
(5) $F1^3$	0.46	0.42	0.28	0.47	1.00		
(6) Market Return	-0.10	0.04	0.09	-0.25	-0.07	1.00	

Figure A1: Factor Interaction Regressions — Intercepts The figure reports the intercept α_i from the individual factor interaction regression $R_{\text{smgd},i} = \alpha_i + \beta_{1i}R_{\text{orig},i} + \beta_{2i}Z_t + \beta_{3i}(R_{\text{orig},i} \times Z_t) + \varepsilon_t$, estimated separately for each of the 18 anomalies and each macroeconomic variable Z . The intercept represents the unconditional alpha of the skewness management return — the premium that remains after accounting for the direct and interactive effects of the macro variable. Cell values show the estimated intercept with significance stars. Colors reflect the t -statistic on a scale clipped at ± 5 . Newey-West HAC standard errors with 12 lags.

